

On $\mathbb{E}_\infty$ - and DAlg-Descendability

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Abstract

For a map $R \rightarrow S$ of $\mathbb{E}_\infty$ -rings, its n th finite Amitsur stage is

$$\mathrm{Tot}^n(S^{\otimes_R(\bullet+1)}).$$

Antieau and Stefanich asked five questions about multiplicative retractions from these finite stages. We answer all five.

For every prime p and every $k \geq 2$, the restriction map

$$C^*(BC_{p^k}; \mathbb{F}_p) \longrightarrow C^*(BC_p; \mathbb{F}_p)$$

is descendable but not $\mathbb{E}_\infty$ -descendable, and hence is not DAlg-descendable; for $k = 1$ the map is the identity. Thus ordinary and $\mathbb{E}_\infty$ -descendability differ even for cochains on finite cyclic p -groups.

Complexification $KO \rightarrow KU$ is $\mathbb{E}_\infty$ -descendable: the unit

$$KO \longrightarrow \mathrm{Tot}^5(KU^{\otimes_{KO}(\bullet+1)})$$

admits a retraction in CAlg_{KO} . The units at stages 0, 1, and 2 do not split in CAlg_{KO} , and the unit at stage 3 does not split even as an $\mathbb{E}_1$ - KO -algebra map. Thus the least $\mathbb{E}_\infty$ -splitting stage is either 4 or 5.

Let A be a Noetherian $\mathbb{F}_p$ -algebra of finite Krull dimension, put $d_A = \max\{1, \dim A\}$, and let $A \rightarrow B$ be a faithfully flat map of discrete $\mathbb{F}_p$ -algebras. Without any finite-type hypothesis on B , the unit admits a retraction in derived commutative A -algebras

$$\mathrm{Tot}^{d_A+1}(B^{\otimes_A(\bullet+1)}) \longrightarrow A;$$

specializing to $\mathbb{F}_p[x] \rightarrow \mathbb{F}_p[x^{1/p^\infty}]$ gives a retraction at stage 2. The proof uses the first two natural layers of a free derived symmetric algebra, a two-term induced resolution over the Frobenius skew-polynomial ring $A[F]$, and integral DAlg flattening.

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1 Introduction

Let $f: R \rightarrow S$ be a map of $\mathbb{E}_\infty$ -rings. Ordinary descendability means that the canonical comparison from the constant pro- R -module $\{R\}$ to the tower

$$\left\{ \mathrm{Tot}^n(S^{\otimes_R(\bullet+1)}) \right\}_n$$

is an equivalence. Antieau and Stefanich study two multiplicative refinements. They call f $\mathbb{E}_\infty$ -descendable when this comparison is an equivalence in pro- $\mathbb{E}_\infty$ -rings. For maps of derived commutative rings, they define DAlg-descendability by requiring the analogous comparison in pro-derived-commutative rings [1]. The implications are

$$\mathrm{DAlg}\text{-descendable} \implies \mathbb{E}_\infty\text{-descendable} \implies \text{descendable}.$$

Their finite-retraction criterion, recalled in Section 2, identifies each of the two pro-conditions with the existence, for some n , of a retraction

$$\mathrm{Tot}^n(S^{\otimes_R(\bullet+1)}) \longrightarrow R$$

in the corresponding algebra category. They ask whether the displayed implications can be strict and whether these finite-stage splittings exist in the following examples.

- (a) Does there exist a descendable map of $\mathbb{E}_\infty$ -rings which is not $\mathbb{E}_\infty$ -descendable?
- (b) Is $KO \rightarrow KU$ $\mathbb{E}_\infty$ -descendable?
- (c) Is $C^*(BC_{p^k}; \mathbb{F}_p) \rightarrow C^*(BC_p; \mathbb{F}_p)$ $\mathbb{E}_\infty$ -descendable or DAlg-descendable?
- (d) Are faithfully flat maps of finite-type $\mathbb{F}_p$ -algebras DAlg-descendable?
- (e) Is $\mathbb{F}_p[x] \rightarrow \mathbb{F}_p[x^{1/p^\infty}]$ DAlg-descendable?

Main results

Questions (a) and (c) have a common answer.

Theorem 1.1 (Cyclic cochains). *Let p be a prime and let $k \geq 2$. For the inclusion $C_p \subset C_{p^k}$, the restriction map*

$$C^*(BC_{p^k}; \mathbb{F}_p) \longrightarrow C^*(BC_p; \mathbb{F}_p)$$

is descendable but not $\mathbb{E}_\infty$ -descendable. It is therefore not DAlg-descendable.

For $k = 1$ the restriction map is the identity. For $k \geq 2$, ordinary descendability is Mathew's [12, Corollary 4.10]; the new content is that no finite stage admits an $\mathbb{E}_\infty$ -retraction. Since every DAlg-retraction gives an $\mathbb{E}_\infty$ -retraction, this also settles the DAlg part of question (c). We model each finite stage by cochains on a space over BC_{p^k} and map that space to a sphere bundle. Mandell's theorem turns a retraction into a section after p -completion, while a nonzero K -theoretic Euler class obstructs such a section.

Question (b) has the opposite outcome: complexification is $\mathbb{E}_\infty$ -descendable, but no $\mathbb{E}_\infty$ -splitting occurs before stage 4.

Theorem 1.2 (Real and complex K-theory). *The complexification map $KO \rightarrow KU$ is $\mathbb{E}_\infty$ -descendable. More precisely, the unit*

$$KO \longrightarrow \mathrm{Tot}^5(KU^{\otimes_{KO}(\bullet+1)})$$

admits a retraction in CAlg_{KO} . The least stage admitting such a retraction is therefore either 4 or 5. In fact, the unit into the third finite Amitsur stage admits no retraction even as an $\mathbb{E}_1$ - KO -algebra map.

For the upper bound, we use the commutative KO -algebras KO/η^k constructed by recursive pullbacks in [9, Definition 3.3(1)]. In Proposition 4.1, we identify KO/η^{n+1} with the n th finite Amitsur stage of $KO \rightarrow KU$. For $n = 5$, its underlying KO -module is $KO \vee \Sigma^7 KO$. After p -completion, the complexification of this stage is $C^*(S^5; KU_p^\wedge)$. The height-one power operation on its top odd-degree class supplies a characteristic equation. We impose this equation by a pushout of free $K(1)$ -local commutative algebras and prove that the pushout remains rank two. At $p = 2$, the required real lift uses Balderrama's equivalence

$$L_{K(1)}((S^{7\sigma_{\Sigma_2}})_{h\Sigma_2}) \simeq S_{K(1)},$$

where σ_{Σ_2} is the sign representation of Σ_2 [3, Lemma 3.1.1]. The prime-local retractions then glue through the arithmetic fracture square.

For the lower bound, Picard groups exclude stages 0, 1, and 2, while the multiplication computed by Levy–Li–Zhang excludes the third stage even as an $\mathbb{E}_1$ - KO -algebra. We do not exclude the fourth stage, so the least splitting stage is either 4 or 5.

Questions (d) and (e) follow from one faithfully flat descent theorem. Let A be a Noetherian $\mathbb{F}_p$ -algebra of finite Krull dimension, set $d_A = \max\{1, \dim A\}$, and let $A \rightarrow B$ be a faithfully flat map of discrete $\mathbb{F}_p$ -algebras. Then

$$A \longrightarrow \mathrm{Tot}^{d_A+1}(B^{\otimes_A(\bullet+1)})$$

has a retraction in DAlg_A . No finite-type hypothesis on B is needed; the precise statement is Theorem 7.1. The proof begins with the flat Frobenius-cell lemma Lemma 7.6. Priddy's operations P^0, P^1 and Petrov's identifications describe the first two layers of the relevant free derived symmetric algebra. A two-term induced resolution over the Frobenius skew-polynomial ring $A[F]$ replaces the potentially nonconvergent monadic bar construction, and Raynaud–Gruson's projective-dimension bound removes the remaining Frobenius obstruction. Integral DAlg flattening then produces the retraction.

For the perfect-closure map $\mathbb{F}_p[x] \rightarrow \mathbb{F}_p[x^{1/p^\infty}]$, the theorem gives the stage-2 retraction required in (e). The unit at stage 0 does not split in $\mathrm{DAlg}_{\mathbb{F}_p[x]}$; whether the unit at stage 1 splits in $\mathrm{DAlg}_{\mathbb{F}_p[x]}$ remains undecided.

Organization. Section 2 recalls the finite-retraction criterion and the punctured-cube model. Section 3 proves Theorem 1.1. Sections 4–6 establish the lower and upper bounds for $KO \rightarrow KU$ and glue the prime-local retractions. Section 7 proves the faithfully flat theorem and applies it to questions (d) and (e).

2 Finite Amitsur stages and retractions

Every argument below uses two facts about the finite partial totalizations of the Amitsur complex: $\mathbb{E}_\infty$ -descendability and DAlg -descendability are each detected by a retraction at one finite stage in the corresponding algebra category, and each stage is computed by a punctured cube. We record both facts and fix the indexing convention.

Proposition 2.1 (Antieau–Stefanich). *Let $f: R \rightarrow S$ be a map of $\mathbb{E}_\infty$ -rings. Then f is $\mathbb{E}_\infty$ -descendable if and only if, for some $n \geq 0$, the unit*

$$R \longrightarrow \mathrm{Tot}^n(S^{\otimes_R(\bullet+1)})$$

admits a retraction in CAlg_R . For maps of derived commutative rings, the identical statement holds with $\mathbb{E}_\infty$ and CAlg_R replaced by DAlg and DAlg_R .

Proof. This is [1, Proposition 2.3 and the preceding DAlg variant]. Write

$$u_n: R \longrightarrow \mathrm{Tot}^n(S^{\otimes_R(\bullet+1)})$$

for the unit. If the comparison from the constant object $\{R\}$ to the tower of finite partial totalizations is a pro-equivalence, represent an inverse by

$$r_n: \mathrm{Tot}^n(S^{\otimes_R(\bullet+1)}) \longrightarrow R$$

at some stage n . The equality with the identity in the pro-category means that, for some $m \geq n$,

$$r_n \circ \left(\mathrm{Tot}^m(S^{\otimes_R(\bullet+1)}) \longrightarrow \mathrm{Tot}^n(S^{\otimes_R(\bullet+1)}) \right) \circ u_m = \mathrm{id}_R.$$

Thus r_n composed with the transition from stage m to stage n is an actual retraction of u_m .

Conversely, a finite-stage retraction places the unit in the finite-limit-and-retract closure of commutative S -algebras, which is the criterion for $\mathbb{E}_\infty$ -descendability in loc. cit. The same argument in DAlg_R proves the stated DAlg variant. $\square$

The punctured-cube model also determines the underlying module of every finite partial totalization.

Lemma 2.2 (Punctured-cube model). *Let $f: R \rightarrow S$ be a map of $\mathbb{E}_\infty$ -rings. The zeroth partial totalization is S , and for every $n \geq 1$ there is a natural equivalence in CAlg_R*

$$\mathrm{Tot}^n(S^{\otimes_R(\bullet+1)}) \simeq \mathrm{Tot}^{n-1}(S^{\otimes_R(\bullet+1)}) \times_{\mathrm{Tot}^{n-1}(S^{\otimes_R(\bullet+1)}) \otimes_R S} S.$$

The underlying R -module of the n th partial totalization is

$$\mathrm{cofib}(\mathrm{fib}(R \rightarrow S)^{\otimes_R(n+1)} \longrightarrow R).$$

The same assertions hold in DAlg_R for a map of derived commutative rings.

Proof. The zeroth partial totalization is the degree-zero term S . For $n \geq 1$, the usual right-cofinal functor from nonempty subsets of an $(n+1)$ -element set to the truncated simplex category identifies the n th partial totalization with the limit of the punctured $(n+1)$ -cube generated by $R \rightarrow S$. We separate this cube according to whether the last coordinate occurs.

The nonempty subsets not containing the last coordinate form the punctured n -cube, whose limit is the $(n-1)$ st partial totalization. The subsets containing the last coordinate form a full n -cube with initial vertex S , so their limit is S . Their common restriction is the punctured n -cube after base change from R to S . Its limit is

$$\mathrm{Tot}^{n-1}(S^{\otimes_R(\bullet+1)}) \otimes_R S.$$

Indeed, the forgetful functor creates limits, while tensoring with S is exact on the stable module category and therefore preserves finite limits of underlying modules. Gluing the two parts of the cube now gives

$$\mathrm{Tot}^{n-1}(S^{\otimes_R(\bullet+1)}) \times_{\mathrm{Tot}^{n-1}(S^{\otimes_R(\bullet+1)}) \otimes_R S} S.$$

This proves the recursive equivalence in the statement. The underlying-module identification is the Adams-tower description [13, Proposition 2.14]. The same cubical argument applies to derived commutative rings. $\square$

Remark 2.3. Thus the zeroth partial totalization is S . If $\mathrm{fib}(R \rightarrow S) \simeq \Sigma^d R$ and the map $\mathrm{fib}(R \rightarrow S)^{\otimes(n+1)} \rightarrow R$ is multiplication by a class v^{n+1} , then the underlying module of the n th partial totalization is R/v^{n+1} . In particular, there is no indexing shift.

3 Cyclic cochains

Fix a prime p and an integer $k \geq 2$. Here $C_p \subset C_{p^k}$ denotes the unique subgroup of order p . The proof has three steps. We first realize every finite Amitsur stage as cochains on a finite skeletal approximation to $BC_p \rightarrow BC_{p^k}$. We then map this approximation, over BC_{p^k} , to a sphere bundle. Finally, Mandell's theorem converts an $\mathbb{E}_\infty$ -retraction into a section after p -completion, while a K -theoretic Euler class rules out such a section.

3.1 A geometric model for every finite stage

Let $E_\bullet(C_{p^k}/C_p)$ be the usual contractible free simplicial (C_{p^k}/C_p) -set, with

$$E_j(C_{p^k}/C_p) = (C_{p^k}/C_p)^{j+1}.$$

Regard it as a C_{p^k} -simplicial set through the quotient map $C_{p^k} \twoheadrightarrow C_{p^k}/C_p$.

Lemma 3.1. *For every $n \geq 0$, there is a natural equivalence of commutative $C^*(BC_{p^k}; \mathbb{F}_p)$ -algebras*

$$\begin{aligned} \mathrm{Tot}^n \left(C^*(BC_p; \mathbb{F}_p)^{\otimes_{C^*(BC_{p^k}; \mathbb{F}_p)}(\bullet+1)} \right) \\ \simeq C^* \left(EC_{p^k} \times_{C_{p^k}} \left| \mathrm{sk}_n E_\bullet(C_{p^k}/C_p) \right|; \mathbb{F}_p \right). \end{aligned}$$

Under this equivalence the unit into the partial totalization is induced by the projection

$$EC_{p^k} \times_{C_{p^k}} \left| \mathrm{sk}_n E_\bullet(C_{p^k}/C_p) \right| \longrightarrow BC_{p^k}.$$

Proof. The proof is degreewise. We first identify tensoring with $C^*(BC_p; \mathbb{F}_p)$ with adjoining the finite set C_{p^k}/C_p . Naturality then assembles these identifications into an equivalence of cosimplicial commutative algebras. Finally, cochains turn the truncated realization into the corresponding partial totalization, and the levelwise projection to a point identifies the unit.

The covering

$$BC_p \simeq EC_{p^k} \times_{C_{p^k}} (C_{p^k}/C_p) \longrightarrow BC_{p^k}$$

is associated with the transitive C_{p^k} -set C_{p^k}/C_p . For every finite C_{p^k} -set U , write

$$C^0(U; \mathbb{F}_p) = \text{Map}_{\text{Set}}(U, \mathbb{F}_p)$$

for the corresponding permutation object in $\text{Fun}(BC_{p^k}, \text{Mod}^\omega(\mathbb{F}_p))$. Since C_{p^k} is a finite p -group, Mathew's symmetric-monoidal equivalence

$$\text{Fun}(BC_{p^k}, \text{Mod}^\omega(\mathbb{F}_p)) \simeq \text{Perf}(C^*(BC_{p^k}; \mathbb{F}_p))$$

[12, Theorem 2.29 and equation (4); compare Proposition 4.7] takes $C^0(U; \mathbb{F}_p)$ to

$$C^0(U; \mathbb{F}_p)^{hC_{p^k}} \simeq C^*(EC_{p^k} \times_{C_{p^k}} U; \mathbb{F}_p).$$

Under this equivalence $C^0(C_{p^k}/C_p; \mathbb{F}_p)$ corresponds to $C^*(BC_p; \mathbb{F}_p)$. Since

$$C^0(C_{p^k}/C_p; \mathbb{F}_p) \otimes_{\mathbb{F}_p} C^0(U; \mathbb{F}_p) \simeq C^0((C_{p^k}/C_p) \times U; \mathbb{F}_p),$$

symmetric monoidality gives a natural equivalence

$$\begin{aligned} C^*(BC_p; \mathbb{F}_p) \otimes_{C^*(BC_{p^k}; \mathbb{F}_p)} C^*(EC_{p^k} \times_{C_{p^k}} U; \mathbb{F}_p) \\ \simeq C^*(EC_{p^k} \times_{C_{p^k}} ((C_{p^k}/C_p) \times U); \mathbb{F}_p). \end{aligned}$$

Taking $U = (C_{p^k}/C_p)^j$ and inducting on j yields, for every $j \geq 0$,

$$\begin{aligned} C^*(BC_p; \mathbb{F}_p)^{\otimes_{C^*(BC_{p^k}; \mathbb{F}_p)} (j+1)} \\ \simeq C^*(EC_{p^k} \times_{C_{p^k}} (C_{p^k}/C_p)^{j+1}; \mathbb{F}_p). \end{aligned}$$

Because the equivalence used above is symmetric monoidal, these are equivalences of commutative $C^*(BC_{p^k}; \mathbb{F}_p)$ -algebras, not merely of underlying modules. The construction is natural for maps of finite C_{p^k} -sets. It therefore identifies the Amitsur cofaces and codegeneracies with those of the simplicial Borel construction in the commutative-algebra category.

The simplicial Borel construction $EC_{p^k} \times_{C_{p^k}} E_\bullet(C_{p^k}/C_p)$ has j -simplices

$$EC_{p^k} \times_{C_{p^k}} (C_{p^k}/C_p)^{j+1}.$$

Here F_{Sp} denotes the function spectrum. Since the cochain functor $C^*(-; \mathbb{F}_p) = F_{\text{Sp}}(\Sigma_+^\infty(-), H\mathbb{F}_p)$ sends homotopy colimits of spaces to homotopy limits of commutative algebras,

$$\begin{aligned} C^*(EC_{p^k} \times_{C_{p^k}} \left| \text{sk}_n E_\bullet(C_{p^k}/C_p) \right|; \mathbb{F}_p) \\ \simeq \lim_{[j] \in \Delta_{\leq n}} C^*(EC_{p^k} \times_{C_{p^k}} (C_{p^k}/C_p)^{j+1}; \mathbb{F}_p) \\ \simeq \text{Tot}^n \left(C^*(BC_p; \mathbb{F}_p)^{\otimes_{C^*(BC_{p^k}; \mathbb{F}_p)} (\bullet+1)} \right). \end{aligned}$$

The projection from this Borel construction to BC_{p^k} is induced levelwise by $(C_{p^k}/C_p)^{j+1} \rightarrow *$, so the corresponding cochain map is the unit into this partial totalization. $\square$

3.2 A sphere-bundle obstruction

Fix n . We map the Borel construction in Lemma 3.1, over BC_{p^k} , to a sphere bundle whose projection has no section after p -completion.

Let χ_{p^k} denote the tautological one-dimensional complex character of C_{p^k} . The character $\chi_{p^k}^p$ has kernel C_p and therefore defines a faithful character of C_{p^k}/C_p . Its $(n+2)$ -fold direct sum has a unit sphere

$$S((\chi_{p^k}^p)^{\oplus(n+2)}),$$

which is a free (C_{p^k}/C_p) -space of dimension $2n+3$.

Lemma 3.2. *There exists a C_{p^k} -equivariant map*

$$|\mathrm{sk}_n E_\bullet(C_{p^k}/C_p)| \longrightarrow S((\chi_{p^k}^p)^{\oplus(n+2)}).$$

Consequently there is a map over BC_{p^k}

$$\begin{aligned} \phi_n^{\mathrm{sph}}: EC_{p^k} \times_{C_{p^k}} |\mathrm{sk}_n E_\bullet(C_{p^k}/C_p)| \\ \longrightarrow EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)}). \end{aligned}$$

Proof. Construct the map cellularly. The free (C_{p^k}/C_p) -CW complex $|\mathrm{sk}_n E_\bullet(C_{p^k}/C_p)|$ has dimension n , whereas

$$S((\chi_{p^k}^p)^{\oplus(n+2)}) \cong S^{2n+3}$$

is $(2n+2)$ -connected. Each orbit of d -cells is $(C_{p^k}/C_p) \times D^d$, so its equivariant extension problem is the ordinary extension problem for a map

$$S^{d-1} \longrightarrow S((\chi_{p^k}^p)^{\oplus(n+2)}).$$

Since $d \leq n$, the obstruction lies in a vanishing homotopy group of this sphere. Starting with one representative from each orbit of zero-cells therefore produces a (C_{p^k}/C_p) -equivariant, hence C_{p^k} -equivariant, map. Applying the Borel construction gives the asserted map over BC_{p^k} . $\square$

A cochain retraction would force such a section.

Lemma 3.3 (Cochain retractions detect sections). *Let $q: Y \rightarrow Z$ be a map of connected nilpotent spaces of finite p -type. If*

$$C^*(Z; \mathbb{F}_p) \longrightarrow C^*(Y; \mathbb{F}_p)$$

admits a retraction as an $\mathbb{E}_\infty$ - $C^(Z; \mathbb{F}_p)$ -algebra, then the completed map*

$$q_p^\wedge: Y_p^\wedge \longrightarrow Z_p^\wedge$$

admits a section.

Proof. Let

$$r: C^*(Y; \mathbb{F}_p) \longrightarrow C^*(Z; \mathbb{F}_p)$$

be the given retraction of q^* . We make two replacements in order to put r in the category to which Mandell's theorem applies: we extend scalars to $\overline{\mathbb{F}_p}$ and replace both spaces by their p -completions.

For $W = Y, Z$, finite p -type gives an equivalence

$$H\overline{\mathbb{F}}_p \otimes_{H\mathbb{F}_p} C^*(W; \mathbb{F}_p) \xrightarrow{\sim} C^*(W; \overline{\mathbb{F}}_p).$$

Indeed, on π_{-j} this is the usual finite-dimensional coefficient-extension isomorphism in $H^j(W; -)$, and both sides are coconnective. Moreover, the completion map $W \rightarrow W_p^\wedge$ is a mod- p equivalence, so it induces an equivalence on cochains with coefficients in $\overline{\mathbb{F}}_p$. After these identifications, scalar extension of r becomes a retraction

$$r_{\overline{\mathbb{F}}_p}: C^*(Y_p^\wedge; \overline{\mathbb{F}}_p) \longrightarrow C^*(Z_p^\wedge; \overline{\mathbb{F}}_p)$$

of $(q_p^\wedge)^*$.

Richter–Shipley provide the comparison between commutative Eilenberg–Mac Lane algebra spectra and the differential graded E_∞ -algebras used in Mandell’s formulation [18]. Mandell’s main theorem, together with its ring-spectrum form in Appendix C, makes the contravariant cochain functor fully faithful on the homotopy category of connected p -complete nilpotent spaces of finite p -type [11, Main Theorem and Appendix C]. Under this fully faithful functor, the algebra map $r_{\overline{\mathbb{F}}_p}$ is represented by a map

$$s: Z_p^\wedge \longrightarrow Y_p^\wedge.$$

Contravariance carries $r_{\overline{\mathbb{F}}_p} \circ (q_p^\wedge)^* = \text{id}$ to $q_p^\wedge \circ s \simeq \text{id}_{Z_p^\wedge}$. Thus s is a section. $\square$

Lemma 3.4 (The sphere bundle is nilpotent of finite p -type). *For every $n \geq 0$, the spaces BC_{p^k} and*

$$EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)})$$

are connected nilpotent spaces of finite p -type. Their p -completion maps are therefore mod- p equivalences.

Proof. The sphere-bundle fibration gives

$$\pi_1(EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)})) \cong C_{p^k}.$$

The scalar action of C_{p^k} on the higher homotopy groups of the odd-dimensional sphere is trivial: it factors through the connected group $U(n+2)$, so every scalar transformation is homotopic to the identity. Thus the total space is nilpotent. Its finite p -type follows from the Serre spectral sequence of the sphere-bundle fibration; the same properties for BC_{p^k} are immediate. The final assertion is the standard completion theorem for connected nilpotent spaces of finite p -type [7]. $\square$

Lemma 3.5 (Euler obstruction). *The map*

$$\pi_n^{\text{sph}}: EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)}) \longrightarrow BC_{p^k}$$

does not admit a section after p -completion.

Proof. The argument uses three cohomology theories for three separate purposes. Periodic K -theory detects that the Euler class is nonzero. A finite p -power quotient of connective K -theory retains a nonzero image of this class but kills it over the sphere bundle. That quotient theory is invariant under p -completion, which turns a hypothetical section into a contradiction.

The K -theoretic Euler class of the complex representation $(\chi_{p^k}^p)^{\oplus(n+2)}$ is

$$e((\chi_{p^k}^p)^{\oplus(n+2)}) = \lambda_{-1}\left((\chi_{p^k}^p)^{\oplus(n+2)}\right)^* = (1 - \chi_{p^k}^{-p})^{n+2} \in KU^0(BC_{p^k}).$$

Let $I_{\text{aug}} \subset R(C_{p^k})$ be the augmentation ideal. By the Atiyah–Segal completion theorem [2],

$$KU^0(BC_{p^k}) \cong R(C_{p^k})_{I_{\text{aug}}}^{\wedge}.$$

Evaluation at a primitive p^k th root of unity defines a continuous homomorphism

$$R(C_{p^k})_{I_{\text{aug}}}^{\wedge} \longrightarrow \mathbb{Z}_p[\zeta_{p^k}], \quad \chi_{p^k} \longmapsto \zeta_{p^k},$$

because $\zeta_{p^k} - 1$ is topologically nilpotent. Since $k \geq 2$, we have $\zeta_{p^k}^p \neq 1$, and hence

$$e((\chi_{p^k}^p)^{\oplus(n+2)}) \longmapsto (1 - \zeta_{p^k}^{-p})^{n+2} \neq 0.$$

Thus this Euler class is nonzero. More precisely, it cannot be divisible by every power of p : otherwise its displayed image would lie in $\bigcap_r p^r \mathbb{Z}_p[\zeta_{p^k}] = 0$, since this finite torsion-free $\mathbb{Z}_p$ -module is p -adically separated.

Since ku is the connective cover of KU , the map $\Omega^\infty ku \rightarrow \Omega^\infty KU$ is an equivalence. Therefore

$$ku^0(X) \xrightarrow{\sim} KU^0(X)$$

for every CW complex X , in particular for BC_{p^k} . Choose $r \geq 1$ such that the reduction of the Euler class

$$\bar{e}_{r,n} \in (ku/p^r)^0(BC_{p^k})$$

is nonzero. Such an r exists by the preceding nondivisibility and the cofiber sequence $ku \xrightarrow{p^r} ku \rightarrow ku/p^r$. Over the unit sphere bundle, the pullback of $(\chi_{p^k}^p)^{\oplus(n+2)}$ has its tautological nowhere-zero section, so its Euler class vanishes:

$$(\pi_n^{\text{sph}})^* \bar{e}_{r,n} = 0 \quad \text{in} \quad (ku/p^r)^0\left(EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)})\right).$$

The spectrum ku/p^r is bounded below, and its homotopy groups are annihilated by p^r , hence are derived p -complete. By [4, Theorem 2.6(2)], ku/p^r is therefore $H\mathbb{F}_p$ -local. By Lemma 3.4, completion consequently induces isomorphisms on $(ku/p^r)^0$ for both the base and the total space. If s were a section of $(\pi_n^{\text{sph}})_p^{\wedge}$, then

$$s^*((\pi_n^{\text{sph}})_p^{\wedge})^* = \text{id},$$

so $((\pi_n^{\text{sph}})_p^{\wedge})^*$ would be injective. Under the completion isomorphisms, however, the nonzero class $\bar{e}_{r,n}$ still pulls back to zero. This is the desired contradiction. $\square$

Proof of Theorem 1.1. Ordinary descendability is [12, Corollary 4.10]. Assume for contradiction that the map is $\mathbb{E}_\infty$ -descendable. By Proposition 2.1, there would exist an n and a retraction in $\text{CAlg}_{C^*(BC_{p^k}; \mathbb{F}_p)}$

$$r: \text{Tot}^n\left(C^*(BC_p; \mathbb{F}_p)^{\otimes_{C^*(BC_{p^k}; \mathbb{F}_p)}(\bullet+1)}\right) \longrightarrow C^*(BC_{p^k}; \mathbb{F}_p).$$

Let ϕ_n^{sph} be the map over BC_{p^k} supplied by Lemma 3.2. Transport r across the equivalence of Lemma 3.1. The composite

$$\begin{aligned} & C^*(EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)}); \mathbb{F}_p) \\ & \xrightarrow{(\phi_n^{\text{sph}})^*} C^*(EC_{p^k} \times_{C_{p^k}} |\text{sk}_n E_\bullet(C_{p^k}/C_p)|; \mathbb{F}_p) \\ & \xrightarrow{r} C^*(BC_{p^k}; \mathbb{F}_p) \end{aligned}$$

is a retraction in $\text{CAlg}_{C^*(BC_{p^k}; \mathbb{F}_p)}$ of the pullback along π_n^{sph} ,

$$C^*(BC_{p^k}; \mathbb{F}_p) \longrightarrow C^*(EC_{p^k} \times_{C_{p^k}} S((\chi_{p^k}^p)^{\oplus(n+2)}); \mathbb{F}_p).$$

Indeed, ϕ_n^{sph} lies over BC_{p^k} , so composing this pullback with the displayed composite gives the identity of $C^*(BC_{p^k}; \mathbb{F}_p)$ by the retraction property of r .

The hypotheses of Lemma 3.3 are verified in Lemma 3.4. Hence the completed map $(\pi_n^{\text{sph}})_p^\wedge$ would admit a section, contradicting Lemma 3.5. Hence the map is not $\mathbb{E}_\infty$ -descendable. Any DAlg-retraction would give an $\mathbb{E}_\infty$ -retraction after applying the forgetful functor $\text{DAlg}_{C^*(BC_{p^k}; \mathbb{F}_p)} \rightarrow \text{CAlg}_{C^*(BC_{p^k}; \mathbb{F}_p)}$, so the map is not DAlg-descendable either. $\square$

Corollary 3.6 (Answers to questions (a) and (c)). *Question (a) of Antieau–Stefanich has a positive answer. In question (c), the map is the identity for $k = 1$ and is neither $\mathbb{E}_\infty$ - nor DAlg-descendable for every $k \geq 2$.*

4 The finite $KO \rightarrow KU$ tower

This section identifies the finite Amitsur stages with the commutative KO -algebras constructed recursively in [9, Definition 3.3(1)] and excludes retractions through stage 3.

Wood’s cofiber sequence

$$\Sigma KO \xrightarrow{\eta} KO \longrightarrow KU$$

identifies the fiber of $KO \rightarrow KU$ with ΣKO , so Lemma 2.2 gives the following description.

Proposition 4.1. *For every $n \geq 0$, there is a natural equivalence in CAlg_{KO}*

$$\text{Tot}^n(KU^{\otimes_{KO}(\bullet+1)}) \simeq KO/\eta^{n+1},$$

where the right-hand side carries the commutative algebra structure constructed by Levy–Li–Zhang [9, Definition 3.3(1)]. Here and below, in an algebraic context, KO/η^{n+1} denotes this specified commutative KO -algebra; we say explicitly when only its underlying module is meant. Its underlying KO -module is the cofiber of multiplication by η^{n+1} . In particular,

$$KO/\eta^6 \simeq KO \vee \Sigma^7 KO$$

as a KO -module.

Proof. Wood’s sequence identifies $\text{fib}(KO \rightarrow KU)$ with ΣKO , and Lemma 2.2 identifies the underlying module of

$$\text{Tot}^n(KU^{\otimes_{KO}(\bullet+1)})$$

with

$$\text{cofib}(\Sigma^{n+1} KO \xrightarrow{\eta^{n+1}} KO) \simeq KO/\eta^{n+1}.$$

Moreover, Lemma 2.2 gives the recursive pullback

$$\mathrm{Tot}^n(KU^{\otimes_{KO}(\bullet+1)}) \simeq \mathrm{Tot}^{n-1}(KU^{\otimes_{KO}(\bullet+1)}) \times_{\mathrm{Tot}^{n-1}(KU^{\otimes_{KO}(\bullet+1)}) \otimes_{KO} KU} KU \quad (n \geq 1).$$

Starting from the zeroth partial totalization KU , this is precisely the recursive pullback construction of the commutative algebra KO/η^{n+1} in [9, Definition 3.3(1)]. Finally, $\eta^3 = 0$ in $\pi_* KO$. Hence, for $n \geq 2$, the map η^{n+1} in the displayed cofiber sequence is null, so its cofiber splits as $KO \vee \Sigma^{n+2} KO$; compare [9, Lemma 3.8]. $\square$

4.1 Low-stage obstructions

Picard groups exclude stages 0, 1, and 2; the multiplication computed by Levy–Li–Zhang excludes stage 3.

Proposition 4.2. *For $0 \leq n \leq 2$, the unit*

$$KO \longrightarrow \mathrm{Tot}^n(KU^{\otimes_{KO}(\bullet+1)})$$

admits no $\mathbb{E}_\infty$ - KO -algebra retraction.

Proof. An $\mathbb{E}_\infty$ -retraction would make the base-change homomorphism

$$\mathrm{Pic}(KO) \longrightarrow \mathrm{Pic}(KO/\eta^{n+1})$$

split injective. The relevant Picard groups are

$$\mathrm{Pic}(KO) = \mathbb{Z}/8, \quad \mathrm{Pic}(KO/\eta) = \mathrm{Pic}(KU) = \mathbb{Z}/2, \quad \mathrm{Pic}(KO/\eta^2) = \mathrm{Pic}(KO/\eta^3) = \mathbb{Z}/4.$$

For the last two computations see [9, Proposition 3.13]. None of the target groups contains a copy of $\mathbb{Z}/8$. $\square$

Proposition 4.3. *The unit $KO \rightarrow KO/\eta^4$, where the target carries the descent-tower algebra structure of Proposition 4.1, has no retraction as an $\mathbb{E}_1$ - KO -algebra.*

Proof. The contradiction uses a class $x \in \pi_5(KO/\eta^4)$ with boundary 1. Its square is $\eta^2\beta \neq 0$, where $\beta \in \pi_8 KO$ is the real Bott class. Any KO -algebra retraction would kill x , since $\pi_5 KO = 0$, and would fix x^2 .

The boundary-one class maps to ηu_2 . The Levy–Li–Zhang multiplication calculation [9, Remark 3.10] gives

$$\pi_*(KO/\eta^3) \cong \mathbb{Z}[\eta, \epsilon, u_2^{\pm 1}]/(2\eta, \eta^3, \epsilon\eta, \epsilon^2), \quad |\eta| = 1, \quad |\epsilon| = 0, \quad |u_2| = 4.$$

Here u_2 is the degree-four periodicity unit and ϵ is the degree-zero square-zero generator used in this presentation. Choose $x \in \pi_5(KO/\eta^4)$ whose boundary in the cofiber sequence

$$\Sigma^4 KO \xrightarrow{\eta^4} KO \longrightarrow KO/\eta^4 \longrightarrow \Sigma^5 KO$$

is $1 \in \pi_0 KO$; such a lift exists because $\eta^4 = 0$ and the cofiber long exact sequence is exact. The tower map

$$KO/\eta^4 \longrightarrow KO/\eta^3$$

is induced by the square whose left vertical map is multiplication by η and whose right vertical map is the identity. Naturality of the boundary therefore shows that the image of x has boundary η . The displayed ring gives

$$\pi_5(KO/\eta^3) \cong \mathbb{Z}/2\{\eta u_2\}.$$

Since $\eta \neq 0$, the image of x is nonzero; as the displayed group has order two, that image is ηu_2 . Therefore x^2 maps to

$$\eta^2 u_2^2 \neq 0.$$

The square is $x^2 = \eta^2 \beta$. To identify x^2 , use the module splitting $KO/\eta^4 \simeq KO \vee \Sigma^5 KO$:

$$\pi_{10}(KO/\eta^4) \cong \pi_{10}KO \oplus \pi_5 KO \cong \mathbb{Z}/2\{\eta^2 \beta\},$$

and the first summand is induced by the unit. Since the image of x^2 in $\pi_{10}(KO/\eta^3)$ is nonzero, x^2 is the nonzero unit class $\eta^2 \beta$.

The square obstructs a retraction. If an $\mathbb{E}_1$ - KO -algebra retraction $r: KO/\eta^4 \rightarrow KO$ existed, then $r(x) = 0$ because $\pi_5 KO = 0$, so $r(x^2) = 0$. On the other hand, KO -linearity forces $r(x^2) = \eta^2 \beta \neq 0$, a contradiction. $\square$

5 Prime-local retractions at the fifth stage

Fix a prime p , and write

$$KO_p^\wedge/\eta^6 := (KO/\eta^6) \otimes_{KO} KO_p^\wedge$$

for the specified commutative algebra carried by the fifth finite Amitsur stage after p -completion. Fix a Bott unit $v \in \pi_2 KU_p^\wedge$. The map $KO_p^\wedge \rightarrow KU_p^\wedge$ is a faithful C_2^{gal} -Galois extension [19, Proposition 5.3.1 and Lemma 7.1.1]. We write $\gamma \in C_2^{\text{gal}}$ for its nontrivial automorphism, complex conjugation, and σ_{gal} for the sign representation of C_2^{gal} . All spectra in this section are implicitly $K(1)$ -local at p .

Proposition 5.1 (Local retraction). *For every prime p , the unit*

$$KO_p^\wedge \longrightarrow KO_p^\wedge/\eta^6$$

admits a retraction in the ordinary category $\text{CAlg}_{KO_p^\wedge}(\text{Sp})$.

We first construct a retraction among $K(1)$ -local commutative algebras. Galois descent identifies the complexified fifth stage with $C^*(S^5; KU_p^\wedge)$; its top class is an eigenvector for the height-one power operation. After lifting the resulting relation to $KO_p^\wedge$, we impose it by a pushout. A rank-two calculation identifies that pushout with $KO_p^\wedge/\eta^6$, so its augmentation gives the local retraction. To promote it to an ordinary commutative-algebra map, it remains to prove that the functor

$$\text{CAlg}_{KO_p^\wedge}(\text{Sp}_{K(1)}) \longrightarrow \text{CAlg}_{KO_p^\wedge}(\text{Sp})$$

is fully faithful; this supplies the retraction asserted above.

5.1 The top class and its power operation

Lemma 5.2. *There is an equivalence of commutative $KU_p^\wedge$ -algebras*

$$\begin{aligned} KU_p^\wedge \otimes_{KO_p^\wedge} (KO_p^\wedge/\eta^6) &\simeq C^*(S^5; KU_p^\wedge) \\ &= F_{\text{Sp}}(S_+^5, KU_p^\wedge). \end{aligned}$$

Moreover, the complexification map

$$KO_p^\wedge/\eta^6 \longrightarrow C^*(S^5; KU_p^\wedge)$$

induces an isomorphism on π_7 . Let $u \in \pi_1 C^*(S^5; KU_p^\wedge)$ be the reduced top generator of S^5 , put $\omega_{\mathbb{C}} = v^3 u \in \pi_7 C^*(S^5; KU_p^\wedge) \cong \mathbb{Z}_p$, and let $\omega_{\mathbb{R}} \in \pi_7(KO_p^\wedge/\eta^6)$ be its unique lift.

Proof. For a faithful finite G -Galois extension $A \rightarrow B$, iteration of $B \otimes_A B \simeq F_{\mathrm{Sp}}(G_+, B)$ gives an equivalence of cosimplicial commutative B -algebras

$$B \otimes_A B^{\otimes_A(\bullet+1)} \simeq F_{\mathrm{Sp}}((G^{\bullet+1})_+, B).$$

Because B is dualizable over A , base change by B preserves the finite limit defining Tot^n . Thus, for every $n \geq 0$,

$$B \otimes_A \mathrm{Tot}^n(B^{\otimes_A(\bullet+1)}) \simeq F_{\mathrm{Sp}}(|\mathrm{sk}_n E_\bullet(G)|_+, B)$$

as commutative B -algebras. Apply this with $A = KO_p^\wedge$, $B = KU_p^\wedge$, and $G = C_2^{\mathrm{gal}}$, and use Proposition 4.1 to identify the fifth stage. The standard C_2^{gal} -equivariant model identifies $|\mathrm{sk}_5 E_\bullet(C_2^{\mathrm{gal}})|$ with $S(6\sigma_{\mathrm{gal}})$. Hence

$$\begin{aligned} KU_p^\wedge \otimes_{KO_p^\wedge} (KO_p^\wedge / \eta^6) &\simeq C^*(|\mathrm{sk}_5 E_\bullet(C_2^{\mathrm{gal}})|; KU_p^\wedge) \\ &\simeq C^*(S(6\sigma_{\mathrm{gal}}); KU_p^\wedge) \\ &\simeq C^*(S^5; KU_p^\wedge). \end{aligned}$$

The last equivalence forgets the C_2^{gal} -action and uses the underlying homeomorphism $S(6\sigma_{\mathrm{gal}}) \cong S^5$.

It remains to identify the top summand. The first of the module splittings

$$\begin{aligned} KO_p^\wedge / \eta^6 &\simeq KO_p^\wedge \vee \Sigma^7 KO_p^\wedge, \\ C^*(S^5; KU_p^\wedge) &\simeq KU_p^\wedge \vee \Sigma^7 KU_p^\wedge. \end{aligned}$$

comes from the cofiber sequence for multiplication by η^6 , which is null, and its base change to $KU_p^\wedge$ gives the second splitting under the equivalence above. Thus the map on the degree-7 summands is the suspension of $\pi_0 KO_p^\wedge \rightarrow \pi_0 KU_p^\wedge$, hence is an isomorphism. $\square$

Let

$$\begin{aligned} \mathcal{F}_p &= \mathrm{Free}_{\mathrm{CAlg}_{KO_p^\wedge}}^{K(1)}(\Sigma^7 KO_p^\wedge), \\ \mathcal{F}_p^\mathbb{C} &= KU_p^\wedge \otimes_{KO_p^\wedge} \mathcal{F}_p. \end{aligned}$$

and let $x_\mathbb{R} \in \pi_7 \mathcal{F}_p$ and $x_\mathbb{C} \in \pi_7 \mathcal{F}_p^\mathbb{C}$ denote the free generators. For $n \geq 0$, write $\mathcal{F}_{p, \langle n \rangle}$ and $(\mathcal{F}_p^\mathbb{C})_{\langle n \rangle}$ for the natural arity- n (equivalently, weight- n) retracts. The angle brackets distinguish arity from localization at a prime.

We now fix the periodic normalization of the odd operation used throughout this section. Let $Q_1: \pi_1 C \rightarrow \pi_1 C$ be the Barthel–Frankland–Hopkins operation on a p -complete H_∞ $KU_p^\wedge$ -algebra C . For every $j \in \mathbb{Z}$ and $z \in \pi_{2j+1} C$, define

$$Q(z) := v^j Q_1(v^{-j} z) \in \pi_{2j+1} C. \quad (5.1)$$

This is natural in C . The $\mathbb{Z}/2$ -periodic free-monad calculation of [5, Proposition 6.5, Corollary 6.6, Lemma 6.13, and Theorem 6.14] supplies the two properties used below. First, on a free odd generator y , the classes $y, Qy, Q^2y, \dots$ are the strictly exterior generators, of weights $1, p, p^2, \dots$. Thus $Q^i y$ has weight p^i , matching the arity decomposition just introduced.

Second, suppose that C carries a compatible semilinear C_2^{gal} -action whose restriction to $KU_p^\wedge$ is complex conjugation, and write γ for its nontrivial element. Then $\gamma(v) = -v$; additivity and the ring-level naturality of Hopkins’s construction [5, Remark 6.9] give

$$\gamma Q(z) = (-1)^j v^j Q_1((-1)^j v^{-j} \gamma z) = Q(\gamma z).$$

Lemma 5.3. *The top class satisfies*

$$Q(\omega_{\mathbb{C}}) = p^2 \omega_{\mathbb{C}} \quad \text{in } \pi_7 C^*(S^5; KU_p^\wedge).$$

Proof. Barthel–Frankland construct the natural degree-one operation Q_1 on the homotopy of p -complete $H_\infty KU_p^\wedge$ -algebras [5, Proposition 6.8], and [5, Remark 6.9] identifies their normalization with Hopkins’s. The odd component of this same operation is denoted θ^1 by Heuts–Land.

Let $u \in \pi_1 C^*(S^5; KU_p^\wedge) = KU_p^{-1}(S^5)$ be the reduced top generator chosen in Lemma 5.2. In the proof of [8, Proposition 4.22], the suspension compatibility $\theta^1 \leftrightarrow \theta^0$ and the calculation $\theta^0(\beta^r) = p^{r-1} \beta^r$ show that

$$Q_1(u) = p^2 u,$$

because $5 = 2r - 1$ with $r = 3$. Here β is the KU -Bott class in the normalization of the cited calculation; our choice of v fixes the periodicity identification in (5.1). Since $\omega_{\mathbb{C}} = v^3 u$, the periodic normalization (5.1) gives

$$Q(\omega_{\mathbb{C}}) = v^3 Q_1(u) = p^2 \omega_{\mathbb{C}}.$$

The scalar p^2 is nonzero in $\mathbb{Z}_p$. This is the complex relation that will be lifted and imposed below. The calculation uses the periodic operation (5.1); it does not transport an unstable power operation merely through a suspension isomorphism of reduced K -cohomology groups. $\square$

To impose the relation $Qx_{\mathbb{C}} = p^2 x_{\mathbb{C}}$ over $KO_p^\wedge$, it remains to lift $Qx_{\mathbb{C}} \in \pi_7 \mathcal{F}_p^{\mathbb{C}}$ to a class of $\mathcal{F}_p$.

Lemma 5.4 (Real lift). *There exists $\widetilde{Q}x_{\mathbb{R}} \in \pi_7 \mathcal{F}_p$ whose image in $\pi_7 \mathcal{F}_p^{\mathbb{C}}$ is $Qx_{\mathbb{C}}$.*

Proof. The two cases use different forms of descent.

At odd primes, $Qx_{\mathbb{C}}$ is Galois fixed. Galois descent and stability of faithful Galois extensions under base change identify $\mathcal{F}_p$ with $(\mathcal{F}_p^{\mathbb{C}})^{hC_2^{\text{gal}}}$ [19, Lemma 7.1.1], and the natural arity, or weight, decomposition is stable under the C_2^{gal} -action. The weight- p line of $\pi_7 \mathcal{F}_p^{\mathbb{C}}$ is free of rank one over $\mathbb{Z}_p$, generated by $Qx_{\mathbb{C}}$; see [5, Proposition 6.5, Corollary 6.6, and Theorem 6.14]. The map $\mathcal{F}_p^{\mathbb{C}} \rightarrow C^*(S^5; KU_p^\wedge)$ classifying $\omega_{\mathbb{C}}$ is the base change of the real map $\mathcal{F}_p \rightarrow KO_p^\wedge/\eta^6$ classifying $\omega_{\mathbb{R}}$, so it is C_2^{gal} -equivariant. On the weight- p line it is multiplication by p^2 :

$$Qx_{\mathbb{C}} \mapsto Q(\omega_{\mathbb{C}}) = p^2 \omega_{\mathbb{C}}.$$

This is injective because $p^2 \neq 0$ in the torsion-free ring $\mathbb{Z}_p$. Since $\omega_{\mathbb{C}}$ and p^2 are fixed, equivariance and injectivity imply that $Qx_{\mathbb{C}}$ is fixed.

At odd primes, the fixed class lifts over $KO_p^\wedge$. Regard $(\mathcal{F}_p^{\mathbb{C}})_{\langle p \rangle}$ as a $KO_p^\wedge$ -module with C_2^{gal} -action. Since p is odd, 2 is invertible in $\pi_0 KO_p^\wedge$, and the idempotents

$$e_+ = \frac{1 + \gamma}{2}, \quad e_- = \frac{1 - \gamma}{2}$$

split the group algebra $KO_p^\wedge[C_2^{\text{gal}}]$, where γ is the nontrivial element of C_2^{gal} . In particular, the trivial $KO_p^\wedge[C_2^{\text{gal}}]$ -module is the projective summand $KO_p^\wedge \simeq e_+ KO_p^\wedge[C_2^{\text{gal}}]$. Hence

averaging gives a spectrum-level equivalence

$$\begin{aligned} ((\mathcal{F}_p^{\mathbb{C}})_{\langle p \rangle})^{hC_2^{\text{gal}}} \\ \simeq F_{KO_p^{\wedge}[C_2^{\text{gal}}]}(KO_p^{\wedge}, (\mathcal{F}_p^{\mathbb{C}})_{\langle p \rangle}) \\ \simeq e_+(\mathcal{F}_p^{\mathbb{C}})_{\langle p \rangle}. \end{aligned}$$

Therefore, for every t , the homotopy groups of the left-hand side identify with the C_2^{gal} -fixed part of $\pi_t((\mathcal{F}_p^{\mathbb{C}})_{\langle p \rangle})$. As the arity decomposition is C_2^{gal} -stable, descent identifies $\mathcal{F}_{p, \langle p \rangle}$ with $((\mathcal{F}_p^{\mathbb{C}})_{\langle p \rangle})^{hC_2^{\text{gal}}}$. The fixed class $Qx_{\mathbb{C}}$ therefore has a lift $\widetilde{Qx}_{\mathbb{R}} \in \pi_7 \mathcal{F}_{p, \langle p \rangle} \subset \pi_7 \mathcal{F}_p$.

At $p = 2$, the real weight-two summand is $\Sigma^7 KO_2^{\wedge}$. Here the preceding averaging argument is unavailable, so we compute the weight-two summand directly. The permutation group Σ_2 in the extended power should not be confused with the Galois group C_2^{gal} . The relative weight-two summand is

$$\mathcal{F}_{2, \langle 2 \rangle} \simeq KO_2^{\wedge} \otimes L_{K(1)} D_2(S^7).$$

Here $D_2(S^7) = (S^7 \wedge S^7)_{h\Sigma_2}$. The transposition representation on $\mathbb{R}^7 \oplus \mathbb{R}^7$ splits as $7 \cdot 1 \oplus 7\sigma_{\Sigma_2}$, where σ_{Σ_2} is the sign representation of the permutation group Σ_2 . Hence

$$D_2(S^7) \simeq \Sigma^7 (S^{7\sigma_{\Sigma_2}})_{h\Sigma_2}.$$

Balderrama proves $L_{K(1)}((S^{7\sigma_{\Sigma_2}})_{h\Sigma_2}) \simeq S_{K(1)}$ [3, Lemma 3.1.1]. Consequently

$$\mathcal{F}_{2, \langle 2 \rangle} \simeq \Sigma^7 KO_2^{\wedge}.$$

At $p = 2$, complexification is an isomorphism on weight-two π_7 . The arity decomposition commutes with base change. Choose the displayed sphere-level equivalence once and tensor it first with $KO_2^{\wedge}$ and then with $KU_2^{\wedge}$. This gives the commutative square

$$\begin{array}{ccc} \pi_7(KO_2^{\wedge} \otimes \Sigma^7 S_{K(1)}) & \longrightarrow & \pi_7(KU_2^{\wedge} \otimes \Sigma^7 S_{K(1)}) \\ \downarrow \cong & & \downarrow \cong \\ \pi_0 KO_2^{\wedge} & \longrightarrow & \pi_0 KU_2^{\wedge}. \end{array}$$

The bottom map is the isomorphism $\mathbb{Z}_2 \rightarrow \mathbb{Z}_2$; hence so is the weight-two base-change map on π_7 .

The prime 2: the prescribed class has a real preimage. By [5, Proposition 6.5, Corollary 6.6, and Theorem 6.14], the primitive generator of the complex weight-two line is $Qx_{\mathbb{C}}$. The same strictly exterior calculation gives $x_{\mathbb{C}}^2 = 0$, so there is no decomposable ambiguity in this identification. No normalization of the chosen equivalence with $S_{K(1)}$ is required: the base-change map is an isomorphism onto the entire weight-two group. It therefore supplies a real preimage $\widetilde{Qx}_{\mathbb{R}}$ of $Qx_{\mathbb{C}}$. $\square$

5.2 The characteristic pushout

The relation map. The goal is to impose the relation $Qx_{\mathbb{C}} = p^2 x_{\mathbb{C}}$ on the free generator. By freeness, there is an $\mathbb{E}_{\infty}$ - $KO_p^{\wedge}$ -algebra endomorphism

$$\rho_{p, \mathbb{R}}^{\text{rel}}: \mathcal{F}_p \longrightarrow \mathcal{F}_p, \quad \rho_{p, \mathbb{R}}^{\text{rel}}(x_{\mathbb{R}}) = \widetilde{Qx}_{\mathbb{R}} - p^2 x_{\mathbb{R}}.$$

Write $\rho_{p, \mathbb{C}}^{\text{rel}}$ for its base change to $\mathcal{F}_p^{\mathbb{C}}$; thus

$$\rho_{p, \mathbb{C}}^{\text{rel}}(x_{\mathbb{C}}) = Qx_{\mathbb{C}} - p^2 x_{\mathbb{C}}.$$

Compatibility with the top class. Let $\text{cl}_{\omega,p}: \mathcal{F}_p \rightarrow KO_p^\wedge/\eta^6$ classify $\omega_{\mathbb{R}}$, and let $\varepsilon_p: \mathcal{F}_p \rightarrow KO_p^\wedge$ be the augmentation. We first compare $\text{cl}_{\omega,p} \circ \rho_{p,\mathbb{R}}^{\text{rel}}$ with the composite $\mathcal{F}_p \xrightarrow{\varepsilon_p} KO_p^\wedge \rightarrow KO_p^\wedge/\eta^6$. After complexification,

$$(\text{cl}_{\omega,p} \circ \rho_{p,\mathbb{R}}^{\text{rel}})(x_{\mathbb{R}}) \mapsto Q(\omega_{\mathbb{C}}) - p^2 \omega_{\mathbb{C}} = 0.$$

Since

$$\pi_7(KO_p^\wedge/\eta^6) \longrightarrow \pi_7 C^*(S^5; KU_p^\wedge)$$

is an isomorphism, $(\text{cl}_{\omega,p} \circ \rho_{p,\mathbb{R}}^{\text{rel}})(x_{\mathbb{R}}) = 0$ already in $\pi_7(KO_p^\wedge/\eta^6)$. The composite through ε_p also sends $x_{\mathbb{R}}$ to zero. Because $\mathcal{F}_p$ is free on $\Sigma^7 KO_p^\wedge$, components of the relevant mapping space are detected by the image of $x_{\mathbb{R}}$ in $\pi_7(KO_p^\wedge/\eta^6)$. Choose a path between these two maps. This path is precisely the compatibility datum needed to map the pushout below to $KO_p^\wedge/\eta^6$.

The pushout and its two structure maps. Now form the characteristic pushout in local commutative $KO_p^\wedge$ -algebras

$$\begin{array}{ccc} \mathcal{F}_p & \xrightarrow{\rho_{p,\mathbb{R}}^{\text{rel}}} & \mathcal{F}_p \\ \varepsilon_p \downarrow & & \downarrow \\ KO_p^\wedge & \longrightarrow & \mathcal{A}_p^{\text{char}}. \end{array}$$

Thus

$$\mathcal{A}_p^{\text{char}} = KO_p^\wedge \otimes_{\mathcal{F}_p, \rho_{p,\mathbb{R}}^{\text{rel}}} \mathcal{F}_p.$$

The chosen path supplies the compatibility needed to combine $\text{cl}_{\omega,p}$ and the unit $KO_p^\wedge \rightarrow KO_p^\wedge/\eta^6$ into a comparison map

$$\text{cmp}_p: \mathcal{A}_p^{\text{char}} \longrightarrow KO_p^\wedge/\eta^6.$$

The augmentations on the two copies of $\mathcal{F}_p$ similarly induce $\varepsilon_p^{\text{char}}: \mathcal{A}_p^{\text{char}} \rightarrow KO_p^\wedge$: indeed, $\varepsilon_p(\widetilde{Qx_{\mathbb{R}}}) = 0$ because $\pi_7 KO_p^\wedge = 0$, and hence $\varepsilon_p \circ \rho_{p,\mathbb{R}}^{\text{rel}} \simeq \varepsilon_p$.

The following calculation shows that, after complexification, this recursive relation leaves a free rank-two module with basis 1 and $x_{\mathbb{C}}$.

Lemma 5.5 (Rank-two calculation). *The comparison map $\text{cmp}_p: \mathcal{A}_p^{\text{char}} \rightarrow KO_p^\wedge/\eta^6$ is an equivalence.*

Proof. Complexification detects the comparison map. Faithfulness of $KO_p^\wedge \rightarrow KU_p^\wedge$ reduces the assertion to complexification. The source copy of $\mathcal{F}_p^{\mathbb{C}}$ acts on $KU_p^\wedge$ through its augmentation, while the target copy becomes the $\mathcal{F}_p^{\mathbb{C}}$ -module $(\mathcal{F}_p^{\mathbb{C}})_{\rho_{p,\mathbb{C}}^{\text{rel}}}$ by restriction along $\rho_{p,\mathbb{C}}^{\text{rel}}$. Then

$$\begin{aligned} KU_p^\wedge \otimes_{KO_p^\wedge} \mathcal{A}_p^{\text{char}} \\ \simeq KU_p^\wedge \otimes_{\mathcal{F}_p^{\mathbb{C}}} (\mathcal{F}_p^{\mathbb{C}})_{\rho_{p,\mathbb{C}}^{\text{rel}}}. \end{aligned}$$

It suffices to prove

$$(\mathcal{F}_p^{\mathbb{C}})_{\rho_{p,\mathbb{C}}^{\text{rel}}} \simeq \mathcal{F}_p^{\mathbb{C}} \vee \Sigma^7 \mathcal{F}_p^{\mathbb{C}}.$$

We establish this modulo p^r by a unit-triangular change of variables and then pass to the inverse limit.

The relation map on generators. Barthel–Frankland compute the homotopy of the free local $KU_p^\wedge$ -algebra on an odd generator. Restoring the 2-periodic grading, their Proposition 6.5, Corollary 6.6, and Theorem 6.14 give

$$\pi_* \mathcal{F}_p^{\mathbb{C}} \cong L_0 \left((KU_p^\wedge)_* \otimes_{\mathbb{Z}_p} \Lambda_{\mathbb{Z}_p}(x_{\mathbb{C}}, Qx_{\mathbb{C}}, Q^2x_{\mathbb{C}}, \dots) \right),$$

where L_0 denotes the zeroth left derived functor of p -adic completion of graded modules. Naturality is [5, Proposition 6.8]. The odd operation is a homomorphism of abelian groups by [5, Lemma 6.13]. The same lemma gives its semilinearity; since the ground θ -ring structure fixes $\mathbb{Z}_p$, for a scalar $a \in \mathbb{Z}_p$ and odd classes y, z we have

$$Q(ay) = aQ(y)$$

and $Q(y+z) = Q(y) + Q(z)$. Hence naturality gives the intermediate identity

$$(\rho_{p,\mathbb{C}}^{\text{rel}})_*(Q^i x_{\mathbb{C}}) = Q^i((\rho_{p,\mathbb{C}}^{\text{rel}})_* x_{\mathbb{C}}) = Q^{i+1} x_{\mathbb{C}} - p^2 Q^i x_{\mathbb{C}} =: \delta_i^Q, \quad i \geq 0.$$

Thus it remains to prove that the completed exterior algebra is free of rank two over $\text{im}((\rho_{p,\mathbb{C}}^{\text{rel}})_*)$, with basis $1, x_{\mathbb{C}}$.

The triangular change of variables is invertible modulo p^r . To pass through L_0 -completion, reduce modulo p^r . For $r \geq 1$, put

$$\Lambda_{p^r} = ((KU_p^\wedge)_*/p^r) \otimes_{\mathbb{Z}/p^r} \Lambda_{\mathbb{Z}/p^r}(x_{\mathbb{C}}, Qx_{\mathbb{C}}, Q^2x_{\mathbb{C}}, \dots),$$

and let

$$\Lambda_{p^r}^{\text{rel}} = ((KU_p^\wedge)_*/p^r) \otimes_{\mathbb{Z}/p^r} \Lambda_{\mathbb{Z}/p^r}(\delta_0^Q, \delta_1^Q, \dots) \subset \Lambda_{p^r}.$$

For every n ,

$$Q^n x_{\mathbb{C}} = p^{2n} x_{\mathbb{C}} + \sum_{j=0}^{n-1} p^{2(n-1-j)} \delta_j^Q.$$

On every finite set of variables the transformation

$$(x_{\mathbb{C}}, Qx_{\mathbb{C}}, \dots, Q^n x_{\mathbb{C}}) \longleftrightarrow (x_{\mathbb{C}}, \delta_0^Q, \dots, \delta_{n-1}^Q)$$

is unit triangular. It is therefore invertible over $\mathbb{Z}/p^r$ for every r ; no division by p^2 is involved. Passing to the union over finite sets of variables gives a compatible decomposition

$$\Lambda_{p^r} \cong \Lambda_{p^r}^{\text{rel}} \oplus \Lambda_{p^r}^{\text{rel}} x_{\mathbb{C}}. \quad (5.2)$$

Here the exterior algebras are strictly graded commutative; in particular, the same argument applies at $p = 2$.

The rank-two splitting survives p -adic completion. The coefficient-reduction maps $\Lambda_{p^{r+1}} \rightarrow \Lambda_{p^r}$ are surjective. Hence the tower is Mittag–Leffler and its Milnor $\varprojlim^1$ term vanishes. This is the inverse-limit argument used in the proof of [5, Corollary 6.6]; equivalently, [5, Proposition A.2] identifies the L_0 -completion of this $\mathbb{Z}_p$ -free algebra with the ordinary inverse limit. Consequently

$$\pi_* \mathcal{F}_p^{\mathbb{C}} \cong \varprojlim_r \Lambda_{p^r}.$$

The map induced by $\rho_{p,\mathbb{C}}^{\text{rel}}$ on Λ_{p^r} is an isomorphism onto $\Lambda_{p^r}^{\text{rel}}$, and these isomorphisms and their inverses commute with coefficient reduction. Therefore

$$\text{im}((\rho_{p,\mathbb{C}}^{\text{rel}})_*) \cong \varprojlim_r \Lambda_{p^r}^{\text{rel}}.$$

Taking inverse limits in (5.2) therefore gives

$$\pi_* \mathcal{F}_p^{\mathbb{C}} \cong \text{im}((\rho_{p,\mathbb{C}}^{\text{rel}})_*) \oplus \text{im}((\rho_{p,\mathbb{C}}^{\text{rel}})_*)x_{\mathbb{C}}.$$

Restriction along $\rho_{p,\mathbb{C}}^{\text{rel}}$ is free of rank two. Since $(\rho_{p,\mathbb{C}}^{\text{rel}})_*$ identifies $\pi_* \mathcal{F}_p^{\mathbb{C}}$ with its image, the formula

$$(a, b) \longmapsto \rho_{p,\mathbb{C}}^{\text{rel}}(a) + \rho_{p,\mathbb{C}}^{\text{rel}}(b)x_{\mathbb{C}}$$

defines an $\mathcal{F}_p^{\mathbb{C}}$ -linear map

$$\mathcal{F}_p^{\mathbb{C}} \vee \Sigma^7 \mathcal{F}_p^{\mathbb{C}} \longrightarrow (\mathcal{F}_p^{\mathbb{C}})_{\rho_{p,\mathbb{C}}^{\text{rel}}}$$

that induces an isomorphism on every homotopy group. It is therefore an equivalence of $\mathcal{F}_p^{\mathbb{C}}$ -modules.

The complexified comparison map is an equivalence. Consequently,

$$KU_p^{\wedge} \otimes_{KO_p^{\wedge}} \mathcal{A}_p^{\text{char}} \simeq KU_p^{\wedge} \vee \Sigma^7 KU_p^{\wedge}.$$

Under this splitting, $KU_p^{\wedge} \otimes_{KO_p^{\wedge}} \text{cmp}_p$ sends the two basis elements to 1 and $\omega_{\mathbb{C}}$. By Lemma 5.2, these form the two summands of $C^*(S^5; KU_p^{\wedge}) \simeq KU_p^{\wedge} \vee \Sigma^7 KU_p^{\wedge}$. Thus the complexification of cmp_p is an equivalence, and faithfulness implies that cmp_p is an equivalence. $\square$

Proof of Proposition 5.1. The augmentation $\varepsilon_p^{\text{char}}: \mathcal{A}_p^{\text{char}} \rightarrow KO_p^{\wedge}$, transported across $\text{cmp}_p: \mathcal{A}_p^{\text{char}} \xrightarrow{\sim} KO_p^{\wedge}/\eta^6$, gives the promised retraction in the $K(1)$ -local category. It remains to show that this local morphism is induced by a morphism in the ordinary category $\text{CAlg}_{KO_p^{\wedge}}(\text{Sp})$. It is enough to prove that

$$\text{CAlg}_{KO_p^{\wedge}}(\text{Sp}_{K(1)}) \longrightarrow \text{CAlg}_{KO_p^{\wedge}}(\text{Sp})$$

is fully faithful.

If f is a $K(1)$ -local equivalence and Z is any spectrum, then

$$K(1) \otimes (f \otimes Z) \simeq (K(1) \otimes f) \otimes Z$$

is an equivalence. Thus $L_{K(1)}$ is compatible with the symmetric monoidal structure in the sense of [10, Example 2.2.1.7]. By [10, Proposition 2.2.1.9], the local category inherits the tensor product $(X, Y) \mapsto L_{K(1)}(X \otimes Y)$, and the full inclusion of the associated infinity-operads induces a fully faithful functor

$$\text{CAlg}(\text{Sp}_{K(1)}) \longrightarrow \text{CAlg}(\text{Sp}).$$

Passing to the undercategory of the local algebra $KO_p^{\wedge}$ gives a fully faithful functor

$$\text{CAlg}_{KO_p^{\wedge}}(\text{Sp}_{K(1)}) \longrightarrow \text{CAlg}_{KO_p^{\wedge}}(\text{Sp}).$$

The spectrum $KU_p^{\wedge}$ is local, and $KO_p^{\wedge} \simeq (KU_p^{\wedge})^{hC_2^{\text{gal}}}$ is local because local objects are closed under limits. The underlying $KO_p^{\wedge}$ -module of $KO_p^{\wedge}/\eta^6$ is $KO_p^{\wedge} \vee \Sigma^7 KO_p^{\wedge}$, so it is local as well; the equivalence cmp_p then also shows that $\mathcal{A}_p^{\text{char}}$ is local. Therefore $\varepsilon_p^{\text{char}}$ and cmp_p , constructed in the local category, determine the same maps uniquely in ordinary $\text{CAlg}_{KO_p^{\wedge}}$. Transporting $\varepsilon_p^{\text{char}}$ across cmp_p gives a genuine commutative $KO_p^{\wedge}$ -algebra map $KO_p^{\wedge}/\eta^6 \rightarrow KO_p^{\wedge}$. Its composite with the unit is $\text{id}_{KO_p^{\wedge}}$ because it is a $KO_p^{\wedge}$ -algebra map. Hence it is the required retraction. $\square$

6 Arithmetic gluing

The prime-complete retractions glue with a rational augmentation across the arithmetic fracture square. Their restrictions to the rational overlap are joined by a path because the augmentation space there is connected.

Write

$$\widehat{KO} = \prod_p KO_p^\wedge, \quad KO_{\mathbb{A}_f} := \widehat{KO} \otimes \mathbb{Q}.$$

The subscript $\mathbb{A}_f$ records the finite-adelic overlap. The arithmetic fracture square is a pullback square of $\mathbb{E}_\infty$ -rings

$$KO \simeq KO_{\mathbb{Q}} \times_{KO_{\mathbb{A}_f}} \widehat{KO}.$$

Indeed, every group $\pi_n KO$ is finitely generated, and applying π_n reduces the assertion to the classical arithmetic pullback square for a finitely generated abelian group. Limits of spectra are detected on homotopy groups, and the forgetful functor from commutative ring spectra creates limits.

Lemma 6.1. *For $C = KO_{\mathbb{Q}}$ and for $C = KO_{\mathbb{A}_f}$, there is an equivalence*

$$(KO/\eta^6) \otimes_{KO} C \simeq \text{Free}_{\text{CAlg}_C}(\Sigma^7 C).$$

In particular, the left-hand algebra admits augmentations to C , and the space of such augmentations is connected.

Proof. Send the free generator to the top summand of $(KO/\eta^6) \otimes_{KO} C$. Over a rational base the free commutative algebra on one odd generator is the exterior algebra, so the source has underlying module $C \vee \Sigma^7 C$. The target has the same underlying module by Proposition 4.1. The map is an isomorphism on the unit summand and, by construction, on the top summand; the forgetful functor detects equivalences, so it is an equivalence. Moreover,

$$\pi_0 \text{Map}_{\text{CAlg}_C}(\text{Free}_{\text{CAlg}_C}(\Sigma^7 C), C) \cong \pi_7 C = 0.$$

Thus the augmentation mapping space has a single connected component. $\square$

Proof of Theorem 1.2. First, the KO -module KO/η^6 is perfect and hence dualizable. Therefore

$$(KO/\eta^6) \otimes_{KO} \widehat{KO} \simeq \prod_p ((KO/\eta^6) \otimes_{KO} KO_p^\wedge),$$

so the retractions of Proposition 5.1 combine to a retraction

$$r_{\widehat{KO}} : (KO/\eta^6) \otimes_{KO} \widehat{KO} \longrightarrow \widehat{KO}.$$

Second, over $KO_{\mathbb{Q}}$ choose an augmentation

$$r_{\mathbb{Q}} : (KO/\eta^6) \otimes_{KO} KO_{\mathbb{Q}} \longrightarrow KO_{\mathbb{Q}}$$

supplied by Lemma 6.1. After base change to $KO_{\mathbb{A}_f}$, the restrictions of $r_{\mathbb{Q}}$ and $r_{\widehat{KO}}$ lie in the same component by the connectedness assertion of that lemma. Choose a path between them.

Finally, the representable functor $\text{Map}_{\text{CAlg}_{KO}}(KO/\eta^6, -)$ preserves limits. Applying it to the arithmetic pullback square says that $\text{Map}_{\text{CAlg}_{KO}}(KO/\eta^6, KO)$ is the homotopy pullback of the two restriction maps

$$\text{Map}_{\text{CAlg}_{KO_{\mathbb{Q}}}}((KO/\eta^6) \otimes_{KO} KO_{\mathbb{Q}}, KO_{\mathbb{Q}}) \longrightarrow \text{Map}_{\text{CAlg}_{KO_{\mathbb{A}_f}}}((KO/\eta^6) \otimes_{KO} KO_{\mathbb{A}_f}, KO_{\mathbb{A}_f})$$

and

$$\mathrm{Map}_{\mathrm{CAlg}_{\widehat{KO}}}((KO/\eta^6) \otimes_{KO} \widehat{KO}, \widehat{KO}) \longrightarrow \mathrm{Map}_{\mathrm{CAlg}_{KO_{\mathbb{A}_f}}}((KO/\eta^6) \otimes_{KO} KO_{\mathbb{A}_f}, KO_{\mathbb{A}_f}).$$

Thus the two augmentations and the chosen path determine an $\mathbb{E}_\infty$ - KO -algebra map

$$KO/\eta^6 \longrightarrow KO.$$

Its composite with the unit is the identity because $\mathrm{Map}_{\mathrm{CAlg}_{KO}}(KO, KO)$ is contractible. Thus it is a retraction, and Proposition 2.1 proves that $KO \rightarrow KU$ is $\mathbb{E}_\infty$ -descendable.

The exclusions in Propositions 4.2 and 4.3 give the lower bound 4, while the retraction just constructed gives the upper bound 5 for the least splitting stage. $\square$

7 Faithfully flat maps in characteristic p

Throughout this section we work with nonconnective derived commutative rings. Tensor products of derived algebras are derived by definition. In module formulas we display $\mathbf{L}$ when no flatness hypothesis identifies the derived tensor product with the ordinary one.

Theorem 7.1 (Faithfully flat DAlg descent). *Let A be a Noetherian $\mathbb{F}_p$ -algebra of finite Krull dimension and set*

$$d_A = \max\{1, \dim A\}.$$

For every faithfully flat map $A \rightarrow B$ of discrete $\mathbb{F}_p$ -algebras, with no finiteness assumption on B , the unit

$$A \longrightarrow \mathrm{Tot}^{d_A+1}(B^{\otimes_A(\bullet+1)})$$

admits a retraction in DAlg_A . Consequently $A \rightarrow B$ is DAlg-descendable.

The cubical model makes the displayed Amitsur stage a derived algebra with exactly one negative homotopy group, and that group is flat. The first two layers of a free derived symmetric algebra, together with a two-term resolution over the Frobenius skew-polynomial ring, compute a free derived algebra whose generator has the prescribed Frobenius action. Raynaud–Gruson’s projective-dimension bound then permits a pushout that kills the unique negative homotopy group while preserving a fixed Tor bound. This pushout may create lower homotopy; successive pushouts along free derived symmetric algebras kill those lower classes and leave again a single flat negative homotopy group. Alternating the two constructions makes every transition on that group zero. The resulting colimit is A , and its inverse gives the required retraction.

Frobenius modules. Let A be a discrete $\mathbb{F}_p$ -algebra. Let $A[F]$ denote the Frobenius skew-polynomial ring, with

$$Fa = a^p F \quad (a \in A),$$

and write

$$\varphi_A^{r,*} M = {}_A A_{\varphi_A^r} \otimes_A M, \quad \varphi_A(a) = a^p.$$

Thus a left $A[F]$ -module is an A -module M together with an A -linear Frobenius action map $F_M: \varphi_A^{1,*} M \rightarrow M$. Mathew–Mondal construct a forgetful functor

$$U_F: \mathrm{DAlg}_A \longrightarrow D(A[F])$$

and a left adjoint $\mathrm{Free}_{A[F]}$ [14, Construction 5.12]. By construction, restricting $U_F(R)$ along $A \rightarrow A[F]$ recovers the underlying A -complex of R . The restriction functor

$D(A[F]) \rightarrow D(A)$ is conservative and t -exact for the standard t -structures. Consequently the homotopy groups and Postnikov truncations of $U_F(R)$ are detected on the underlying A -complex.

Tor amplitude. We use the convention that $\mathrm{TorAmp}_A X \leq c$ means $X \otimes_A^{\mathbf{L}} M \in D(A)_{\leq c}$ for every $M \in D(A)_{\leq 0}$. When A is discrete, it is enough to test discrete A -modules: bounded Postnikov truncations are obtained from their homotopy modules by extensions, and an arbitrary coconnective complex is their filtered colimit. Derived tensor product preserves both operations.

Lemma 7.2 (Change of rings for Tor amplitude). *Let C be a derived commutative A -algebra such that*

$$\mathrm{TorAmp}_A \mathrm{cofib}(A \rightarrow C) \leq -1.$$

For every C -module M and every $c \in \mathbb{Z}$,

$$\mathrm{TorAmp}_C M \leq c \iff \mathrm{TorAmp}_A M \leq c,$$

where the term on the right refers to the underlying A -module.

Proof. Apply [14, Proposition 2.14] to the C -module $M[-c]$. $\square$

Ordinary cells. For a flat A -module V and $m \geq 2$, we call the canonical augmented A -algebra $\mathrm{LSym}_A(V[-m])$ an ordinary m -cell.

7.1 The first two layers of an ordinary cell

Only the vanishing range, the first two negative layers, and the Tor amplitude of an ordinary cell enter the argument. In the Frobenius-cell calculation below, degree $-m$ will identify the desired cokernel with the prescribed module, while injectivity in degree $-m - 1$ will exclude an additional leading class. The Tor bound is what allows the resulting comparison fiber to be base changed.

Lemma 7.3 (Natural two-layer calculation). *Let V be a flat A -module and $m \geq 2$. Then $\mathrm{LSym}_A(V[-m])$ is coconnective,*

$$\pi_0 \mathrm{LSym}_A(V[-m]) = A, \quad \pi_{-i} \mathrm{LSym}_A(V[-m]) = 0 \quad (0 < i < m).$$

Write $\iota_V: V \rightarrow \pi_{-m} \mathrm{LSym}_A(V[-m])$ for the weight-one class. There are natural A -linear maps

$$\theta_V^0: \bigoplus_{r \geq 0} \varphi_A^{r,*} V \longrightarrow \pi_{-m} \mathrm{LSym}_A(V[-m]), \quad (7.1)$$

$$\theta_V^1: \bigoplus_{r \geq 0} \varphi_A^{r+1,*} V \longrightarrow \pi_{-m-1} \mathrm{LSym}_A(V[-m]). \quad (7.2)$$

On their r th summands they are represented respectively by

$$(P^0)^r(v) \quad \text{and} \quad P^1(P^0)^r(v),$$

where P^0 is internal Frobenius and P^1 is the unary Priddy operation lowering homotopy degree by one. Both maps are isomorphisms. The augmentation fiber satisfies

$$\mathrm{TorAmp}_A \mathrm{fib}(\mathrm{LSym}_A(V[-m]) \rightarrow A) \leq -m.$$

Proof. The two layer formulas amount to proving that θ_V^0 and θ_V^1 are isomorphisms. We first justify their natural construction, prove the claim over $\mathbb{F}_p$ in rank one, and pass through finite direct sums, base change, and Lazard’s filtered-colimit description of flat modules. The final, independent estimate proves the asserted bound for the augmentation fiber.

The natural comparison maps. Petrov’s Yoneda construction [15, Lemma 3.13] gives natural additive operations P^0 and P^1 on the homotopy of derived commutative $\mathbb{F}_p$ -algebras. Under the comparison between coconnective derived algebras and cosimplicial rings [14, Corollary 3.7], [15, Proposition 3.15(2),(3)] identifies P^0 with internal Frobenius and P^1 with the connecting operation for the termwise exact sequence of underlying cosimplicial abelian groups, for every cosimplicial $\mathbb{F}_p$ -algebra C ,

$$0 \longrightarrow C \xrightarrow{V_{\text{Witt}}} W_2(C) \longrightarrow C \longrightarrow 0.$$

Thus

$$(P^0)^r(av) = a^{p^r}(P^0)^r(v), \quad P^1(P^0)^r(av) = a^{p^{r+1}}P^1(P^0)^r(v).$$

For the second identity, lift multiplication by a through its Teichmüller representative and use $[a]V_{\text{Witt}}(y) = V_{\text{Witt}}(a^py)$. Consequently the formulas

$$\begin{aligned} \theta_{r,V}^0(a \otimes v) &= a(P^0)^r \iota_V(v), \\ \theta_{r,V}^1(a \otimes v) &= aP^1(P^0)^r \iota_V(v) \end{aligned}$$

define natural A -linear maps

$$\begin{aligned} \theta_V^0 &: \bigoplus_{r \geq 0} \varphi_A^{r,*} V \longrightarrow \pi_{-m} \text{LSym}_A(V[-m]), \\ \theta_V^1 &: \bigoplus_{r \geq 0} \varphi_A^{r+1,*} V \longrightarrow \pi_{-m-1} \text{LSym}_A(V[-m]). \end{aligned}$$

The maps θ_V^0 and θ_V^1 are isomorphisms. First suppose that $A = \mathbb{F}_p$ and $V = \mathbb{F}_p$. Priddy’s admissible-monomial calculation [16, Theorem 4.1], in the strict formulation [6, Proposition 7.14 and Remark 7.15], has unary generators indexed by admissible sequences $I = (i_1, \dots, i_s)$ of nonpositive integers, in degree

$$-m + i_1 + \dots + i_s.$$

Every nonconstant generator has degree at most $-m$, so a product of two such generators has degree at most $-2m \leq -m - 2$. In degree $-m$, admissibility forces $I = (0, \dots, 0)$; in degree $-m - 1$, it forces $I = (-1, 0, \dots, 0)$. The latter is admissible for $m \geq 2$: the excess inequality is $-1 > -(p-1)m$ for odd p and $-1 \geq -m$ for $p = 2$. These are precisely the Priddy monomials $(P^0)^r \iota$ and $P^1(P^0)^r \iota$.

Their nonvanishing in the strict derived symmetric power follows by iterating Petrov’s extended-to-strict power comparison in the successive weights p^r [15, Lemma 3.17]; under the Priddy indexing, the resulting composites are exactly the two displayed admissible monomials. Since the corresponding line in each relevant weight is one-dimensional, θ_V^0 and θ_V^1 are isomorphisms. The Brantner–Mathew decomposition is used here only to count the bigraded dimensions. No basis-dependent identification is retained, in accordance with [6, Warning 7.16].

For $V = \mathbb{F}_p^d$, use

$$\text{LSym}_{\mathbb{F}_p}((V \oplus W)[-m]) \simeq \text{LSym}_{\mathbb{F}_p}(V[-m]) \otimes_{\mathbb{F}_p} \text{LSym}_{\mathbb{F}_p}(W[-m]).$$

The Künneth formula and $-2m \leq -m - 2$ show that in the two displayed degrees exactly one tensor factor can be nonconstant. Hence the same two maps are isomorphisms.

If $V \simeq A^d$, base change gives

$$\mathrm{LSym}_A((A^d)[-m]) \simeq A \otimes_{\mathbb{F}_p} \mathrm{LSym}_{\mathbb{F}_p}((\mathbb{F}_p^d)[-m]),$$

and the maps above are the corresponding base-changed maps. They were defined without a basis, and the semilinearity formulas show explicitly that they commute with arbitrary matrices.

For an arbitrary flat A -module V , Lazard's theorem writes V as a filtered colimit of finite free modules. The functors LSym_A and $\varphi_A^{r,*}$, as well as homotopy groups, commute with these filtered colimits. This proves the two natural isomorphisms and the asserted vanishing range for every flat V .

The augmentation fiber has Tor amplitude at most $-m$. It remains to prove that $\mathrm{fib}(\mathrm{LSym}_A(V[-m]) \rightarrow A)[m]$ has Tor amplitude at most zero. By [14, Proposition 2.13], the full subcategory $D(A)_{\mathrm{Tor} \leq 0}$ of objects of Tor amplitude at most zero is the smallest subcategory of $D(A)$ closed under extensions and filtered colimits and containing $A[-i]$ for $i \geq 0$. For $V = A^d$, base change gives

$$\mathrm{fib}(\mathrm{LSym}_A((A^d)[-m]) \rightarrow A)[m] \simeq A \otimes_{\mathbb{F}_p} \mathrm{fib}(\mathrm{LSym}_{\mathbb{F}_p}((\mathbb{F}_p^d)[-m]) \rightarrow \mathbb{F}_p)[m].$$

The right-hand complex before base change is coconnective by the calculation above. Over the field $\mathbb{F}_p$ this is equivalent to Tor amplitude at most zero, so Proposition 2.13 places it in $D(\mathbb{F}_p)_{\mathrm{Tor} \leq 0}$; base change carries the specified generators, extensions, and filtered colimits into $D(A)_{\mathrm{Tor} \leq 0}$. Thus $\mathrm{fib}(\mathrm{LSym}_A((A^d)[-m]) \rightarrow A)[m] \in D(A)_{\mathrm{Tor} \leq 0}$. For a flat $V = \mathrm{colim}_\lambda V_\lambda$, exactness of filtered colimits gives

$$\mathrm{fib}(\mathrm{LSym}_A(V[-m]) \rightarrow A)[m] \simeq \mathrm{colim}_\lambda \mathrm{fib}(\mathrm{LSym}_A(V_\lambda[-m]) \rightarrow A)[m] \in D(A)_{\mathrm{Tor} \leq 0}.$$

Therefore $\mathrm{TorAmp}_A \mathrm{fib}(\mathrm{LSym}_A(V[-m]) \rightarrow A) \leq -m$. $\square$

7.2 Flat Frobenius cells

For $N \in D(A[F])$ and $m \geq 2$, we call $\mathrm{Free}_{A[F]}(N[-m])$ the Frobenius m -cell on N . When N is discrete and its underlying A -module is flat, forgetting the $A[F]$ -action and forming the ordinary cell produces the free Frobenius orbit $\bigoplus_{r \geq 0} \varphi_A^{r,*} N$ in degree $-m$. Under these hypotheses, the Frobenius cell instead imposes $P^0(n) = F_N(1 \otimes n)$, leaving N with its prescribed $A[F]$ -action in the first negative layer. The following two-term resolution realizes these relations by a single pushout in DAlg_A .

Lemma 7.4 (Tensor-algebra resolution). *For every discrete left $A[F]$ -module N , there is a natural exact sequence of left $A[F]$ -modules*

$$0 \longrightarrow A[F] \otimes_A \varphi_A^{1,*} N \xrightarrow{\Delta} A[F] \otimes_A N \longrightarrow N \longrightarrow 0, \quad (7.3)$$

where, for $\tilde{n} = 1 \otimes n \in \varphi_A^{1,*} N$,

$$\Delta(s \otimes \tilde{n}) = sF \otimes n - s \otimes F_N(\tilde{n}).$$

If N is flat over A , then so is $\varphi_A^{1,*} N$.

Proof. This is the relative bar resolution for a module over the tensor algebra $A[F] = T_A({}_A A_{\varphi_A})$. Filter by F -word length. The first term of Δ raises word length and the second does not, so the highest word-length component of a nonzero element cannot vanish. This proves injectivity, and the action map identifies the cokernel with N . Finally, $\varphi_A^{1,*}$ preserves flat modules without any flatness assumption on Frobenius: by Lazard's theorem, write N as a filtered colimit of finite free A -modules and apply ${}_A A_{\varphi_A} \otimes_A -$; the resulting terms are finite free for the left A -module structure. $\square$

Lemma 7.5 (Frobenius base change). *For a map $A \rightarrow A'$ of $\mathbb{F}_p$ -algebras there is a natural isomorphism of $(A', A[F])$ -bimodules*

$$A'[F] \cong A' \otimes_A A[F].$$

For every $N \in D(A[F])$ and every $m \in \mathbb{Z}$, consequently

$$A'[F] \otimes_{A[F]}^{\mathbf{L}} N \simeq A' \otimes_A^{\mathbf{L}} N$$

and

$$\mathrm{Free}_{A[F]}(N[-m]) \otimes_A A' \simeq \mathrm{Free}_{A'[F]}((A'[F] \otimes_{A[F]}^{\mathbf{L}} N)[-m]).$$

Proof. Both bimodules are left A' -free on $\{F^r\}_{r \geq 0}$, and right multiplication by $a \in A$ on the r th summand is multiplication by a^{p^r} . The derived tensor formula follows. For $R \in \mathrm{DAlg}_{A'}$, induction–restriction and the defining adjunction give

$$\begin{aligned} \mathrm{Map}_{\mathrm{DAlg}_{A'}}(A' \otimes_A \mathrm{Free}_{A[F]}(N[-m]), R) &\simeq \mathrm{Map}_{D(A[F])}(N[-m], \mathrm{Res}_{A[F]}^{A'[F]} U_F(R)) \\ &\simeq \mathrm{Map}_{D(A'[F])}((A'[F] \otimes_{A[F]}^{\mathbf{L}} N)[-m], U_F(R)). \end{aligned}$$

By Yoneda, this is the asserted base-change equivalence for Frobenius cells. $\square$

Lemma 7.6 (Flat Frobenius-cell lemma). *Suppose that N is discrete, its underlying A -module is flat, and $m \geq 2$. Then $\mathrm{Free}_{A[F]}(N[-m])$ is coconnective,*

$$\begin{aligned} \pi_0 \mathrm{Free}_{A[F]}(N[-m]) &= A, \\ \pi_{-i} \mathrm{Free}_{A[F]}(N[-m]) &= 0 \quad (0 < i < m), \\ \pi_{-m} \mathrm{Free}_{A[F]}(N[-m]) &\cong N \end{aligned}$$

as Frobenius modules. The augmentation fiber satisfies

$$\mathrm{TorAmp}_A \mathrm{fib}(\mathrm{Free}_{A[F]}(N[-m]) \rightarrow A) \leq -m.$$

Proof. Set

$$\mathrm{Fib}_{\mathrm{unit}}^{A[F]}(m, N) := \mathrm{fib}_{D(A[F])}(U_F(A) \rightarrow U_F(\mathrm{Free}_{A[F]}(N[-m])),$$

and write $\mathrm{Fib}_{\mathrm{unit}}^A(m, N)$ for its underlying A -complex. Thus

$$\mathrm{Fib}_{\mathrm{unit}}^A(m, N) = \mathrm{fib}_{D(A)}(A \rightarrow \mathrm{Free}_{A[F]}(N[-m])).$$

Since the augmentation splits the unit,

$$\mathrm{fib}(\mathrm{Free}_{A[F]}(N[-m]) \rightarrow A) \simeq \mathrm{Fib}_{\mathrm{unit}}^A(m, N)[1].$$

It is therefore enough to prove

$$\begin{aligned} \mathrm{Fib}_{\mathrm{unit}}^A(m, N) &\in D(A)_{\leq -m-1}, & \mathrm{TorAmp}_A \mathrm{Fib}_{\mathrm{unit}}^A(m, N) &\leq -m-1, \\ \pi_{-m-1} \mathrm{Fib}_{\mathrm{unit}}^{A[F]}(m, N) &\cong N \quad \text{as a left } A[F]\text{-module.} \end{aligned}$$

We first establish the corresponding statements for an ordinary-cell comparison fiber, and then compare it with $\mathrm{Fib}_{\mathrm{unit}}^A(m, N)$. More precisely, the two-layer calculation and the resolution identify the leading class of the ordinary comparison fiber with N . A square-zero base-change test gives its Tor bound. Change of rings then transports both conclusions to the unit fiber, and the final fiber sequence shifts them into the three assertions above.

The relation pushout. Let

$$\alpha: \mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \longrightarrow \mathrm{LSym}_A(N[-m])$$

be the ordinary-cell map induced by the first arrow in (7.3). Define

$$\mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N) := \mathrm{fib}_{D(A[F])}(U_F(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])) \rightarrow U_F(\mathrm{LSym}_A(N[-m]))),$$

and write $\mathrm{Fib}_{\mathrm{rel}}^A(m, N)$ for its underlying A -complex. Thus

$$\mathrm{Fib}_{\mathrm{rel}}^A(m, N) = \mathrm{fib}_{D(A)}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow \mathrm{LSym}_A(N[-m])).$$

The exact sequence (7.3), shifted by $-m$, is a cofiber sequence in $D(A[F])$. Together with the zero module, it is therefore a pushout square. Applying the colimit-preserving left adjoint $\mathrm{Free}_{A[F]}$ gives a pushout in DAlg_A . Here $\mathrm{Free}_{A[F]}(0) = A$, and induction–restriction followed by the ordinary free-algebra adjunction gives, for every A -module V and every $R \in \mathrm{DAlg}_A$,

$$\begin{aligned} \mathrm{Map}_{\mathrm{DAlg}_A}(\mathrm{Free}_{A[F]}(A[F] \otimes_A^{\mathbf{L}} V[-m]), R) \\ \simeq \mathrm{Map}_{D(A[F])}(A[F] \otimes_A^{\mathbf{L}} V[-m], U_F(R)) \\ \simeq \mathrm{Map}_{D(A)}(V[-m], U_F(R)) \simeq \mathrm{Map}_{\mathrm{DAlg}_A}(\mathrm{LSym}_A(V[-m]), R). \end{aligned}$$

In the $D(A)$ -mapping space, $U_F(R)$ denotes its underlying A -complex. Yoneda therefore identifies the image of the cofiber square with

$$\mathrm{Free}_{A[F]}(N[-m]) \simeq \mathrm{LSym}_A(N[-m]) \otimes_{\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])} A. \quad (7.4)$$

Thus (7.4) is a pushout of derived commutative algebras, not merely of underlying modules. Since base change is exact,

$$\mathrm{Fib}_{\mathrm{unit}}^A(m, N) \simeq \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_{\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])} A.$$

The ordinary comparison fiber has leading layer N . On $\tilde{n} = 1 \otimes n \in \varphi_A^{1,*} N$, the map α is represented in the first negative layer by $P^0(n) - F_N(\tilde{n})$. In the two layers of Lemma 7.3 it is

$$(P^0)^r(\tilde{n}) \longmapsto (P^0)^{r+1}(n) - (P^0)^r(F_N(\tilde{n})), \quad (7.5)$$

$$P^1(P^0)^r(\tilde{n}) \longmapsto P^1(P^0)^{r+1}(n) - P^1(P^0)^r(F_N(\tilde{n})). \quad (7.6)$$

Since P^0 is internal Frobenius, Lemma 7.3 refines in degree $-m$ to a natural isomorphism of left $A[F]$ -modules

$$\theta_V^0: A[F] \otimes_A V \xrightarrow{\sim} \pi_{-m} U_F(\mathrm{LSym}_A(V[-m])), \quad F^r \otimes v \mapsto (P^0)^r(v).$$

For $V = \varphi_A^{1,*} N$, the square

$$\begin{array}{ccc} A[F] \otimes_A \varphi_A^{1,*} N & \xrightarrow{\Delta} & A[F] \otimes_A N \\ \downarrow \theta_{\varphi_A^{1,*} N}^0 & & \downarrow \theta_N^0 \\ \pi_{-m} U_F(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])) & \xrightarrow{\pi_{-m} U_F(\alpha)} & \pi_{-m} U_F(\mathrm{LSym}_A(N[-m])) \end{array}$$

commutes. Indeed, on $F^r \otimes \tilde{n}$ its upper map is

$$F^{r+1} \otimes n - F^r \otimes F_N(\tilde{n}),$$

which is exactly (7.5) under the vertical maps. Hence Lemma 7.4 identifies the cokernel of $\pi_{-m} U_F(\alpha)$ with N in the abelian category of left $A[F]$ -modules, not merely after restriction to A .

The unit-triangular formula (7.6) is injective after restriction to A , hence injective as a map of $A[F]$ -modules. Direct sums cause no convergence issue because every element has finite support. The relevant part of the long exact sequence is

$$\begin{aligned} \pi_{-m} U_F(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])) &\longrightarrow \pi_{-m} U_F(\mathrm{LSym}_A(N[-m])) \\ &\longrightarrow \pi_{-m-1} \mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N) \\ &\longrightarrow \pi_{-m-1} U_F(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])) \\ &\longrightarrow \pi_{-m-1} U_F(\mathrm{LSym}_A(N[-m])). \end{aligned}$$

The first arrow is injective with cokernel N by the commutative square and Lemma 7.4; the last arrow is injective by (7.6). Exactness therefore gives

$$\pi_{-m-1} \mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N) \cong N$$

as left $A[F]$ -modules. The exact restriction functor $D(A[F]) \rightarrow D(A)$ identifies the underlying A -module of $\mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N)$ with $\mathrm{Fib}_{\mathrm{rel}}^A(m, N)$. Together with the vanishing range, the same exact sequence shows

$$\mathrm{Fib}_{\mathrm{rel}}^A(m, N) \in D(A)_{\leq -m-1}, \quad \pi_{-m-1} \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \cong N. \quad (7.7)$$

The ordinary comparison fiber has Tor amplitude at most $-m-1$. We use the discrete-module test for Tor amplitude stated at the beginning of the section. Given a discrete A -module M , base change to the split square-zero extension $A' = A \oplus M$, and put

$$N' = A'[F] \otimes_{A[F]}^{\mathbf{L}} N \simeq A' \otimes_A N.$$

The last equivalence is Lemma 7.5. Since N is A -flat, N' is a discrete flat A' -module. Tensoring (7.3) with $A'[F]$ identifies it with the canonical two-term resolution of N' from Lemma 7.4; in particular, both induced terms remain discrete and flat. The canonical equivalence

$$A' \otimes_A^{\mathbf{L}} \varphi_A^{1,*} N \simeq \varphi_{A'}^{1,*} (A' \otimes_A N)$$

intertwines the two differentials Δ . Under the base-change equivalences for derived symmetric algebras, the map $\alpha \otimes_A A'$ is exactly the ordinary-cell map induced by this resolution. Exactness of derived base change therefore gives

$$\mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_A^{\mathbf{L}} A' \simeq \mathrm{Fib}_{\mathrm{rel}}^{A'}(m, N').$$

Here the right-hand side denotes the preceding construction with (A, N) replaced by (A', N') . The leading-layer calculation above, now over A' , gives $\mathrm{Fib}_{\mathrm{rel}}^{A'}(m, N') \in D(A')_{\leq -m-1}$. After restriction to A ,

$$\mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_A^{\mathbf{L}} A' \simeq \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \oplus (\mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_A^{\mathbf{L}} M).$$

The second summand therefore lies in $D(A)_{\leq -m-1}$ for every discrete M , which gives

$$\mathrm{TorAmp}_A \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \leq -m - 1. \quad (7.8)$$

The unit fiber has Tor amplitude at most $-m - 1$. By Lemma 7.3,

$$\mathrm{TorAmp}_A \mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow A) \leq -m.$$

Since the augmentation splits the unit,

$$\mathrm{cofib}(A \rightarrow \mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])) \simeq \mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow A).$$

The cofiber in the preceding display has A -Tor amplitude at most $-m \leq -1$ by Lemma 7.3. Applying Lemma 7.2 with $c = -m - 1$ to (7.8) gives

$$\mathrm{TorAmp}_{\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])} \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \leq -m - 1.$$

This is the estimate needed to base change to A . Indeed,

$$\mathrm{Fib}_{\mathrm{unit}}^A(m, N) \simeq \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_{\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])} A,$$

so

$$\mathrm{TorAmp}_A \mathrm{Fib}_{\mathrm{unit}}^A(m, N) \leq -m - 1.$$

Base change preserves the leading layer. A second application of Lemma 7.2, now with $c = -m$ and the module $\mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow A)$, gives

$$\mathrm{TorAmp}_{\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])} \mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow A) \leq -m.$$

Tensoring the fiber sequence

$$\mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow A) \longrightarrow \mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \longrightarrow A$$

with $\mathrm{Fib}_{\mathrm{rel}}^A(m, N)$ gives

$$\begin{aligned} & \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_{\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m])} \mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*} N)[-m]) \rightarrow A) \\ & \longrightarrow \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \longrightarrow \mathrm{Fib}_{\mathrm{unit}}^A(m, N). \end{aligned}$$

The first term has Tor amplitude at most $-2m - 1$, so $\mathrm{Fib}_{\mathrm{rel}}^A(m, N) \rightarrow \mathrm{Fib}_{\mathrm{unit}}^A(m, N)$ is an isomorphism on π_{-m-1} .

The leading class has the prescribed Frobenius action. Apply U_F to the pushout square (7.4) and take horizontal fibers in the stable category $D(A[F])$. This produces a morphism of fiber sequences

$$\begin{array}{ccccc} \mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N) & \longrightarrow & U_F(\mathrm{LSym}_A((\varphi_A^{1,*}N)[-m])) & \longrightarrow & U_F(\mathrm{LSym}_A(N[-m])) \\ \downarrow \gamma_{\mathrm{bc}} & & \downarrow & & \downarrow \\ \mathrm{Fib}_{\mathrm{unit}}^{A[F]}(m, N) & \longrightarrow & U_F(A) & \longrightarrow & U_F(\mathrm{Free}_{A[F]}(N[-m])). \end{array}$$

Restriction along $A \rightarrow A[F]$ is exact and identifies the underlying A -modules of $\mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N)$ and $\mathrm{Fib}_{\mathrm{unit}}^{A[F]}(m, N)$ with $\mathrm{Fib}_{\mathrm{rel}}^A(m, N)$ and $\mathrm{Fib}_{\mathrm{unit}}^A(m, N)$, respectively. Under the natural equivalence

$$\mathrm{Fib}_{\mathrm{unit}}^A(m, N) \simeq \mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_{\mathrm{LSym}_A((\varphi_A^{1,*}N)[-m])} A,$$

the underlying map of γ_{bc} is the base-change map. This follows by tensoring the fiber sequence that defines $\mathrm{Fib}_{\mathrm{rel}}^A(m, N)$ over $\mathrm{LSym}_A((\varphi_A^{1,*}N)[-m])$ with A and using (7.4).

The fiber of this underlying map is

$$\mathrm{Fib}_{\mathrm{rel}}^A(m, N) \otimes_{\mathrm{LSym}_A((\varphi_A^{1,*}N)[-m])} \mathrm{fib}(\mathrm{LSym}_A((\varphi_A^{1,*}N)[-m]) \rightarrow A),$$

whose Tor amplitude is at most $-2m - 1$. Thus $\pi_{-m-1}(\gamma_{\mathrm{bc}})$ is an isomorphism. Because γ_{bc} is a morphism in $D(A[F])$, this is an isomorphism of left $A[F]$ -modules.

Since $U_F(A)$ is discrete, the lower fiber sequence gives an $A[F]$ -linear connecting isomorphism

$$\pi_{-m}U_F(\mathrm{Free}_{A[F]}(N[-m])) \xrightarrow{\sim} \pi_{-m-1}\mathrm{Fib}_{\mathrm{unit}}^{A[F]}(m, N).$$

Consequently

$$\pi_{-m}U_F(\mathrm{Free}_{A[F]}(N[-m])) \cong \pi_{-m-1}\mathrm{Fib}_{\mathrm{unit}}^{A[F]}(m, N) \cong \pi_{-m-1}\mathrm{Fib}_{\mathrm{rel}}^{A[F]}(m, N) \cong N$$

through isomorphisms of left $A[F]$ -modules. This is the prescribed Frobenius action on $\pi_{-m}\mathrm{Free}_{A[F]}(N[-m])$.

Finally, after restriction to A , the fiber sequence

$$\mathrm{Fib}_{\mathrm{unit}}^A(m, N) \longrightarrow A \longrightarrow \mathrm{Free}_{A[F]}(N[-m])$$

and $\mathrm{Fib}_{\mathrm{unit}}^A(m, N) \in D(A)_{\leq -m-1}$ give the asserted vanishing range. The equivalence

$$\mathrm{fib}(\mathrm{Free}_{A[F]}(N[-m]) \rightarrow A) \simeq \mathrm{Fib}_{\mathrm{unit}}^A(m, N)[1]$$

and the Tor-amplitude bound on $\mathrm{Fib}_{\mathrm{unit}}^A(m, N)$ give

$$\mathrm{TorAmp}_A \mathrm{fib}(\mathrm{Free}_{A[F]}(N[-m]) \rightarrow A) \leq -m.$$

Thus the fiber sequence proves the asserted vanishing and Tor bound, while the $A[F]$ -linear identification of the leading class proves the remaining Frobenius-module assertion. $\square$

7.3 Projective dimension and Frobenius killing

We first prove the projective-dimension estimate that will make the relevant Postnikov obstruction vanish.

Lemma 7.7 (Projective dimension over $A[F]$). *Let A be Noetherian of finite Krull dimension and put*

$$d_A = \max\{1, \dim A\}.$$

If N is a left $A[F]$ -module which is flat over A , then

$$\mathrm{pd}_{A[F]} N \leq d_A + 1.$$

Proof. Raynaud–Gruson [17, Corollaire 3.2.7] proves that every flat A -module has projective dimension at most d_A . If V is A -flat, apply $A[F] \otimes_A -$ to a projective resolution of V of length d_A . It remains exact because $\mathrm{Tor}_i^A(A[F], V) = 0$ for $i > 0$, and its terms are projective over $A[F]$. Thus

$$\mathrm{pd}_{A[F]}(A[F] \otimes_A V) \leq d_A.$$

Both N and $\varphi_A^{1,*}N$ are A -flat. Hence the two induced modules in

$$0 \longrightarrow A[F] \otimes_A \varphi_A^{1,*}N \longrightarrow A[F] \otimes_A N \longrightarrow N \longrightarrow 0$$

have projective dimension at most d_A . Dimension shifting in this exact sequence gives $\mathrm{pd}_{A[F]} N \leq d_A + 1$. $\square$

For the remainder of the section, assume that A is Noetherian of finite Krull dimension and set

$$d_A = \max\{1, \dim A\}, \quad m = d_A + 1, \quad q = m + 1 = d_A + 2.$$

We use both unit fibers and augmentation fibers below. If an augmentation $R \rightarrow A$ splits the unit $A \rightarrow R$, stability gives

$$\mathrm{fib}(A \rightarrow R) \simeq \mathrm{fib}(R \rightarrow A)[-1].$$

Thus a class in degree $-m$ of an augmented algebra is recorded in degree $-q = -m - 1$ of its unit fiber; this is the only shift encoded by the two parameters.

Definition 7.8 (The two classes of algebras). A map of derived algebras $A \rightarrow R$ is *q -Tor-coconnective* if

$$\mathrm{TorAmp}_A \mathrm{fib}(A \rightarrow R) \leq -q.$$

In particular, a q -Tor-coconnective map is an isomorphism on homotopy in degrees strictly greater than $-q + 1$:

$$\pi_0 R = A, \quad \pi_i R = 0 \quad (-q + 1 < i < 0).$$

It is *q -flat two-stage* if, in addition,

$$\pi_i R = 0 \quad (i \neq 0, -q + 1), \quad \pi_0 R = A, \quad \pi_{-q+1} R \text{ is flat over } A.$$

Here $-q + 1 = -m$. Equivalently, a q -flat two-stage algebra satisfies

$$\mathrm{fib}(A \rightarrow R) \simeq N[-q]$$

for the flat A -module $N = \pi_{-q+1} R$.

The two operations below fit into

$$q\text{-flat two-stage} \xrightarrow{\text{Frobenius killing}} q\text{-Tor-coconnective} \xrightarrow{\text{integral flattening}} q\text{-flat two-stage}.$$

Each pushout used below will be shown to preserve the q -Tor-coconnective condition.

For a q -flat two-stage map $A \rightarrow R$, let $N = \pi_{-m}U_F(R)$ as a left $A[F]$ -module; its underlying A -module is $\pi_{-m}R$. To kill this homotopy group, we first construct a map $\text{Free}_{A[F]}(N[-m]) \rightarrow R$. By adjunction, it is enough to split the two-stage Postnikov extension of $U_F(R)$ in $D(A[F])$. The obstruction is an Ext class over $A[F]$, which Lemma 7.7 forces to vanish.

Lemma 7.9 (Frobenius killing). *Let $A \rightarrow R$ be q -flat two-stage, and let $N = \pi_{-m}U_F(R)$ as a left $A[F]$ -module. There is a map $R \rightarrow R^{\text{kill}_F}$ such that $A \rightarrow R^{\text{kill}_F}$ remains q -Tor-coconnective and*

$$\pi_{-m}R \longrightarrow \pi_{-m}R^{\text{kill}_F}$$

is zero.

Proof. The Postnikov class κ vanishes. Because R is q -flat two-stage, the cofiber of $U_F(A) \rightarrow U_F(R)$ becomes $N[-m]$ after restriction to $D(A)$. The conservativity and t -exactness recorded above identify it with $N[-m]$ in $D(A[F])$, with the induced Frobenius action. Thus the Postnikov triangle of $U_F(R)$ is

$$U_F(A) \longrightarrow U_F(R) \longrightarrow N[-m] \xrightarrow{\kappa} U_F(A)[1].$$

A lift $N[-m] \rightarrow U_F(R)$ inducing the identity on N is a section of the Postnikov projection. Such a section exists precisely when

$$\kappa \in \text{Ext}_{A[F]}^{m+1}(N, U_F(A))$$

vanishes. This follows from Lemma 7.7, since $\text{pd}_{A[F]} N \leq d_A + 1 = m$. Fix such a section.

The Frobenius cell maps onto the leading class. By the defining adjunction of $\text{Free}_{A[F]}(N[-m])$, the chosen section corresponds to a map $g_N: \text{Free}_{A[F]}(N[-m]) \rightarrow R$ inducing the identity on π_{-m} , and we set

$$R^{\text{kill}_F} = R \otimes_{\text{Free}_{A[F]}(N[-m])} A.$$

We must prove that $\text{fib}(g_N)$ has Tor amplitude at most $-q$ over $\text{Free}_{A[F]}(N[-m])$, so that base change to A preserves this bound, and then check that the pushout kills the leading class.

$\text{TorAmp}_A \text{fib}(g_N) \leq -q$. For a discrete A -module M , let $A' = A \oplus M$ be the split square-zero extension, and put $R' = R \otimes_A A'$ and

$$N' = A'[F] \otimes_{A[F]}^{\mathbf{L}} N \simeq A' \otimes_A^{\mathbf{L}} N,$$

with its induced $A'[F]$ -action. Exactness of derived base change and the defining fiber of a q -flat two-stage algebra give

$$\text{fib}(A' \rightarrow R') \simeq \text{fib}(A \rightarrow R) \otimes_A^{\mathbf{L}} A' \simeq N'[-q].$$

The module N' is flat over A' , so R' is again q -flat two-stage. By Lemma 7.5, the base change of the source of g_N is $\text{Free}_{A'[F]}(N'[-m])$. The map $g_N \otimes_A A'$ induces on π_{-m} the base change of id_N , hence $\text{id}_{N'}$. The source and target therefore have the same homotopy

in degrees 0 and $-m$ by Lemma 7.6, and the base-changed map is an isomorphism in both degrees. Thus

$$\mathrm{fib}(g_N) \otimes_A^{\mathbf{L}} A' \simeq \mathrm{fib}(g_N \otimes_A A') \in D(A')_{\leq -m-1} = D(A')_{\leq -q}.$$

After restriction to A ,

$$\mathrm{fib}(g_N) \otimes_A^{\mathbf{L}} A' \simeq \mathrm{fib}(g_N) \oplus (\mathrm{fib}(g_N) \otimes_A^{\mathbf{L}} M).$$

Since this holds for every discrete M , $\mathrm{TorAmp}_A \mathrm{fib}(g_N) \leq -q$.

The killing pushout remains q -Tor-coconnective. The augmentation of $\mathrm{Free}_{A[F]}(N[-m])$ splits its unit, so Lemma 7.6 identifies the A -Tor amplitude of $\mathrm{cofib}(A \rightarrow \mathrm{Free}_{A[F]}(N[-m]))$ as at most $-m \leq -1$. Therefore Lemma 7.2, applied with $c = -q$, upgrades the A -linear estimate to $\mathrm{TorAmp}_{\mathrm{Free}_{A[F]}(N[-m])} \mathrm{fib}(g_N) \leq -q$. Base change gives

$$\mathrm{fib}(A \rightarrow R^{\mathrm{kill}_F}) \simeq \mathrm{fib}(g_N) \otimes_{\mathrm{Free}_{A[F]}(N[-m])} A,$$

so $\mathrm{TorAmp}_A \mathrm{fib}(A \rightarrow R^{\mathrm{kill}_F}) \leq -q$. Thus $A \rightarrow R^{\mathrm{kill}_F}$ is q -Tor-coconnective.

$\pi_{-m} R \rightarrow \pi_{-m} R^{\mathrm{kill}_F}$ is zero. The map $\mathrm{Free}_{A[F]}(N[-m]) \rightarrow R$ is surjective on π_{-m} because it induces the identity on the copy of N . In the composite

$$\mathrm{Free}_{A[F]}(N[-m]) \longrightarrow R \longrightarrow R^{\mathrm{kill}_F} = R \otimes_{\mathrm{Free}_{A[F]}(N[-m])} A,$$

the second description of the pushout identifies this composite with $\mathrm{Free}_{A[F]}(N[-m]) \rightarrow A \rightarrow R^{\mathrm{kill}_F}$, which is zero on negative homotopy. Hence $\pi_{-m} R \rightarrow \pi_{-m} R^{\mathrm{kill}_F}$ is zero. $\square$

7.4 Ordinary cells restore the flat two-stage condition

Frobenius killing preserves the q -Tor-coconnective condition but may create homotopy in degrees at most $-q$. Ordinary cells remove this lower homotopy without changing the Tor-amplitude bound. We first treat one attachment and then iterate it.

Lemma 7.10 (Integral Tor-coconnective cell lemma). *Let $A \rightarrow R$ be q -Tor-coconnective, let P be a flat A -module, and let $k \geq q$. Choose a map*

$$\mathrm{LSym}_A(P[-k]) \longrightarrow R.$$

Then $R' = R \otimes_{\mathrm{LSym}_A(P[-k])} A$ is again q -Tor-coconnective.

Proof. We first estimate the fiber of $\mathrm{LSym}_A(P[-k]) \rightarrow R$ over A . By Lemma 7.3,

$$\mathrm{TorAmp}_A \mathrm{fib}(A \rightarrow \mathrm{LSym}_A(P[-k])) \leq -k - 1 \leq -q - 1, \quad \mathrm{TorAmp}_A \mathrm{fib}(A \rightarrow R) \leq -q.$$

The fiber-comparison triangle

$$\mathrm{fib}(A \rightarrow \mathrm{LSym}_A(P[-k])) \longrightarrow \mathrm{fib}(A \rightarrow R) \longrightarrow \mathrm{fib}(\mathrm{LSym}_A(P[-k]) \rightarrow R)$$

gives $\mathrm{TorAmp}_A \mathrm{fib}(\mathrm{LSym}_A(P[-k]) \rightarrow R) \leq -q$.

To base change along $\mathrm{LSym}_A(P[-k]) \rightarrow A$, we need the same estimate over $\mathrm{LSym}_A(P[-k])$. Here

$$\mathrm{TorAmp}_A \mathrm{cofib}(A \rightarrow \mathrm{LSym}_A(P[-k])) \leq -k \leq -1,$$

so Lemma 7.2, applied with $c = -q$, upgrades the same bound to

$$\mathrm{TorAmp}_{\mathrm{LSym}_A(P[-k])} \mathrm{fib}(\mathrm{LSym}_A(P[-k]) \rightarrow R) \leq -q.$$

Finally, the pushout is base change along $\mathrm{LSym}_A(P[-k]) \rightarrow A$, and

$$\mathrm{fib}(A \rightarrow R') \simeq \mathrm{fib}(\mathrm{LSym}_A(P[-k]) \rightarrow R) \otimes_{\mathrm{LSym}_A(P[-k])} A$$

has Tor amplitude at most $-q$. This is exactly the assertion that R' is q -Tor-coconnective. $\square$

Lemma 7.11 (Integral flattening). *Every q -Tor-coconnective A -algebra R admits a map $R \rightarrow R'$ such that $A \rightarrow R'$ is q -flat two-stage.*

Proof. One round preserves the bound and kills inherited classes. For each $k \geq q$, choose a free A -module $P_k \twoheadrightarrow \pi_{-k}R$. Set $R_q^{\mathrm{att}} = R$. Inductively, the chosen classes and the map $R \rightarrow R_k^{\mathrm{att}}$ give

$$\mathrm{LSym}_A(P_k[-k]) \longrightarrow R \longrightarrow R_k^{\mathrm{att}},$$

and we define

$$R_{k+1}^{\mathrm{att}} = R_k^{\mathrm{att}} \otimes_{\mathrm{LSym}_A(P_k[-k])} A.$$

For these chosen attachments, write

$$R^{\mathrm{round}} := \mathrm{colim}_{k \geq q} R_k^{\mathrm{att}}$$

for the endpoint of the round. This notation records a chosen endpoint; no functorial choice of the surjections $P_k \twoheadrightarrow \pi_{-k}R$ is asserted.

Every successor stage is q -Tor-coconnective by Lemma 7.10. To treat the endpoint, note that filtered colimits in DAlg_A are created on underlying A -modules and are exact there. Consequently

$$\mathrm{fib}(A \rightarrow \mathrm{colim}_{\alpha} R_{\alpha}) \simeq \mathrm{colim}_{\alpha} \mathrm{fib}(A \rightarrow R_{\alpha}).$$

Tensor products commute with these colimits, and $D(A)_{\leq -q}$ is closed under them. Hence R^{round} is also q -Tor-coconnective. At the attachment in degree $-k$, the composite

$$\mathrm{LSym}_A(P_k[-k]) \longrightarrow R_k^{\mathrm{att}} \longrightarrow R_{k+1}^{\mathrm{att}}$$

factors through the augmentation to A . It is therefore zero on π_{-k} . Since $P_k \twoheadrightarrow \pi_{-k}R \rightarrow \pi_{-k}R_k^{\mathrm{att}}$ contains the image of every class originating in $\pi_{-k}R$, the attachment kills all such images. Thus every class in $\pi_i R$ with $i \leq -q$ maps to zero in R^{round} . Classes created during the round need not yet vanish.

Repeated rounds kill all degrees at most $-q$. Starting from $R_0^{\mathrm{flat}} = R$, choose successively one such round beginning at R_n^{flat} and denote its endpoint by R_{n+1}^{flat} . Put

$$R_{\infty}^{\mathrm{flat}} = \mathrm{colim}_n R_n^{\mathrm{flat}}.$$

Each R_n^{flat} , and therefore $R_{\infty}^{\mathrm{flat}}$, is q -Tor-coconnective by the preceding filtered-colimit argument. Let $i \leq -q$ and $x \in \pi_i R_{\infty}^{\mathrm{flat}}$. Since homotopy groups commute with filtered colimits, x is represented by some $x_n \in \pi_i R_n^{\mathrm{flat}}$. The next round kills the image of x_n , so $x = 0$. Thus

$$\pi_i R_{\infty}^{\mathrm{flat}} = 0 \quad (i \leq -q).$$

Only a flat class in degree $-q + 1$ remains. The q -Tor-coconnective condition gives

$$\pi_i R_\infty^{\text{flat}} = 0 \quad (i > 0), \quad \pi_i R_\infty^{\text{flat}} = 0 \quad (-q + 1 < i < 0), \quad \pi_0 R_\infty^{\text{flat}} = A.$$

Together with the preceding vanishing, this leaves only $\pi_{-q+1} R_\infty^{\text{flat}}$ in negative degrees, and

$$\text{fib}(A \rightarrow R_\infty^{\text{flat}}) \simeq \pi_{-q+1} R_\infty^{\text{flat}}[-q].$$

For an A -module M , the condition $\text{TorAmp}_A(M[-q]) \leq -q$ is equivalent to flatness of M . The q -Tor-coconnective bound therefore shows that $\pi_{-q+1} R_\infty^{\text{flat}}$ is flat. Hence R_∞^{flat} is q -flat two-stage. Taking $R' = R_\infty^{\text{flat}}$ proves the lemma. $\square$

7.5 The descent retraction

Proof of Theorem 7.1. The initial flat two-stage algebra. Faithful flatness makes $A \rightarrow B$ pure, hence injective. Its cokernel B/A is flat: for every A -module M ,

$$\text{Tor}_1^A(B/A, M) \cong \ker(M \rightarrow M \otimes_A B) = 0,$$

and $\text{fib}(A \rightarrow B) \simeq (B/A)[-1]$. For every $r \geq 0$, the finite Amitsur identity Lemma 2.2 gives

$$\text{fib}\left(A \rightarrow \text{Tot}^r(B^{\otimes_A(\bullet+1)})\right) \simeq \text{fib}(A \rightarrow B)^{\otimes_A(r+1)} \simeq (B/A)^{\otimes_A(r+1)}[-r-1].$$

Therefore, with $m = d_A + 1$ and $q = m + 1$,

$$\text{fib}\left(A \rightarrow \text{Tot}^m(B^{\otimes_A(\bullet+1)})\right) \simeq (B/A)^{\otimes_A(m+1)}[-m-1].$$

The module $(B/A)^{\otimes_A(m+1)}$ is flat, and the displayed fiber is its shift by $-q$. Hence

$$R_0 = \text{Tot}^m(B^{\otimes_A(\bullet+1)})$$

is q -flat two-stage, with

$$\pi_{-m} R_0 \cong (B/A)^{\otimes_A(m+1)}.$$

Each alternating transition is zero on π_{-m} . Starting with a q -flat two-stage R_i , the two operations give

$$\begin{array}{ccccc} R_i & \longrightarrow & R_i^{\text{kill}_F} & \longrightarrow & R_{i+1} \\ q\text{-flat two-stage} & & q\text{-Tor-coconnective} & & q\text{-flat two-stage.} \end{array}$$

The first arrow is supplied by Lemma 7.9 and satisfies

$$\pi_{-m} R_i \longrightarrow \pi_{-m} R_i^{\text{kill}_F}$$

is zero. The second arrow is supplied by Lemma 7.11. Hence the composite

$$\pi_{-m} R_i \longrightarrow \pi_{-m} R_{i+1}$$

is zero.

The colimit is A . In the sequence

$$R_0 \longrightarrow R_1 \longrightarrow R_2 \longrightarrow \cdots$$

each R_i is q -flat two-stage, so its only possible negative homotopy group is $\pi_{-m}R_i$. The transition maps on these groups are zero, whereas every $\pi_0 R_i$ is A and every transition on π_0 is the identity. Since homotopy groups commute with filtered colimits,

$$\pi_j \left(\operatorname{colim}_i R_i \right) \cong \operatorname{colim}_i \pi_j R_i.$$

For $j = -m$ the right side is the colimit of a system with zero transition maps, hence is zero. For $j = 0$ it is A , and in all other degrees it vanishes. Thus the canonical map

$$A \longrightarrow \operatorname{colim}_i R_i$$

is an equivalence on underlying complexes and therefore in DAlg_A . Composing the map from the initial stage with its inverse gives

$$R_0 \longrightarrow \operatorname{colim}_i R_i \xrightarrow{\sim} A.$$

This is an A -algebra retraction of the unit $A \rightarrow R_0$. Since $m = d_A + 1$, the DAlg finite-retraction criterion Proposition 2.1 (cf. [1, Proposition 2.3]) implies that $A \rightarrow B$ is DAlg -descendable. $\square$

Corollary 7.12 (Answers to questions (d) and (e)). *Every faithfully flat map between finite-type $\mathbb{F}_p$ -algebras is DAlg -descendable. More generally, if the source is any Noetherian finite-dimensional $\mathbb{F}_p$ -algebra, the faithfully flat target need not be of finite type.*

For the perfect-closure map $\mathbb{F}_p[x] \rightarrow \mathbb{F}_p[x^{1/p^\infty}]$, the unit

$$\mathbb{F}_p[x] \longrightarrow \operatorname{Tot}^2 \left(\mathbb{F}_p[x^{1/p^\infty}]^{\otimes_{\mathbb{F}_p[x]} (\bullet+1)} \right)$$

has a retraction in $\operatorname{DAlg}_{\mathbb{F}_p[x]}$. The zeroth stage has no retraction, so the least possible stage is 1 or 2.

Proof. The general assertion follows from Theorem 7.1. For the perfect-closure map, every nonnegative exponent in $\mathbb{Z}[1/p]$ has a unique expression $n + c$ with $n \in \mathbb{Z}_{\geq 0}$ and $c \in [0, 1) \cap \mathbb{Z}[1/p]$. Grouping monomials by this fractional part gives an isomorphism of $\mathbb{F}_p[x]$ -modules

$$\mathbb{F}_p[x^{1/p^\infty}] \cong \bigoplus_{c \in [0, 1) \cap \mathbb{Z}[1/p]} \mathbb{F}_p[x] x^c.$$

Thus $\mathbb{F}_p[x^{1/p^\infty}]$ is free over $\mathbb{F}_p[x]$, and the summand indexed by $c = 0$ shows that the perfect-closure map is faithfully flat. Since $\dim \mathbb{F}_p[x] = 1$, Theorem 7.1 supplies the stage-2 retraction.

On the other hand, an $\mathbb{F}_p[x]$ -algebra map $\mathbb{F}_p[x^{1/p^\infty}] \rightarrow \mathbb{F}_p[x]$ would send $x^{1/p}$ to a polynomial y satisfying $y^p = x$, which is impossible in $\mathbb{F}_p[x]$. Thus the zeroth stage $\mathbb{F}_p[x^{1/p^\infty}]$ has no retraction. We do not decide the stage-1 case. $\square$

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